CE-903 Group Project

Final Project Report

Cross-Domain Automatic Speech Recognition: A Comparative Study of Deep Learning Architectures on Multilingual Datasets

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Abstract

Automatic Speech Recognition (ASR) has become a core building block of modern human-computer interaction systems, but deploying robust ASR across various acoustic domains and accented speech still represents an important open problem. Due to vocabulary mismatch, speaker imbalance and domain shift existing deep learning based automatic speech recognition (ASR) pipelines trained on relatively clean, read-speech corpora like LibriSpeech usually perform significantly worse when evaluated on spontaneous accented or crowd sourced spoken datasets like the Mozilla Common Voice Australian English dataset (en-AU). We conduct this work as a systematic comparison of several state-of-the-art ASR architectures, Whisper-Small, XLS-R (300M), Conformer-CTC, and QuartzNet 5×5, across full fine-tuning and Parameter-Efficient Fine-Tuning (PEFT) setups with Low-Rank Adaptation (LoRA). We employ a two-stage cross-domain training regime, where we first pre-train on LibriSpeech Dataset to learn general acoustic representations, and use target domain adaptation on Common Voice Dataset. Thorough Exploratory Data Analysis (EDA) covering both the quantification of vocabulary shift across datasets, speaker imbalance (with Gini coefficients 0.0966 and 0.8108 respectively), as well as audio quality traits in order to help inform preprocessing and sampling strategies. Experimental results show that LoRA-augmented Whisper-Small reaches 13% WER and 6% CER with only 4.7M trainable parameters, reducing the number of parameters updated during full fine-tuning by a staggering 98.1%, whereas the fully fine-tuned Whisper-Small achieves best WER and CER scores of 9% and 3%, respectively, on our target domain. With the metadata and seven PEFTs, an Omnilingual PEFT architecture extends this to seven languages collectively achieving a sub-5% CER on all evaluated languages using the FLEURS benchmark. Our findings support the efficacy of parameter-efficient adaptation strategies as a compelling and computationally feasible approach for cross-domain, multilingual ASR deployment on consumer-grade hardware.

Index Terms — Automatic Speech Recognition, Deep Learning, LoRA, Parameter-Efficient Fine-Tuning, Whisper, Conformer, XLS-R, Cross-Domain Adaptation, Australian English, Word Error Rate.

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1. Introduction

Automatic Speech Recognition (ASR) technology is a basic component of contemporary human–computer interaction systems that allows machines to transform voice into text. Deep learning has dramatically improved the quality, robustness, and scalability of ASR systems in recent years by using neural architectures (at various levels of processing) such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Transformers and self-supervised representation learning methods. As a result, such deep learning–based ASR systems have become integral to applications including virtual assistants, transcription platforms, accessibility technologies, and large-scale speech analytics.

This report introduces the full project analysis of deep learning based techniques on Automatic Speech Recognition(ASR). This system has been developed in the context of academic research and experimentation; its development focuses on reproducibility, quantitative evaluation, and modular design. An SRS with functional and non-functional requirements, system definition, and a high-level description of operation guide design, development, testing and validation activities.

1.1 Purpose

The current document serves the purpose of detailing (in formally defined terms) the requirements and limitations of a deep learning–based automatic speech recognition system that we designed and built as part of an academic group project. It is also a landmark proposition for students, supervisors and assessors: It articulates precisely what the system should do, how it should do it and when that performance takes place. Specifically, this document aims to:

- In the earlier stages, map out succinct functional/non-functional requirements of ASR system.
- Build a shared understanding, as well as agreement over system goals, assumptions, and constraints.
- Support tracing impact chain between requirements, design decision and evaluation result.
- Facilitate staged development, testing, and validation that are aligned with academic performance assessment and dissertation standards.

1.2 Scope

The suggested system aims to produce an entire end-to-end ASR pipeline that makes use of contemporary deep learning. The system is specialized in transforming speech recordings into regular texts based on some public-available datasets of transcribed speech and neural network–based acoustic models.

The system consists of:

- Ingesting and preprocessing (resampling, normalisation, augmentation etc.) of speech audio data.
- Training and fine-tuning of deep learning ASR models using supervised or self-supervised learning–based methods.
- Text Recognition: transcribing decoded speech to text.
- Quantitative evaluation using Word Error Rate (WER) and Character Error Rate (CER).
- Metrics and ablation with respect to model architecture, preprocessing paradigm, and decoding strategy

We omit commercial deployment, real time delay and high value of (sensitive or private speech) data from the scope. This system is for research, experimentation and academic demonstration purposes only.

1.3 Overview

We describe an end-to-end deep learning based pipeline for Automatic Speech Recognition (ASR) focusing on reproducible cross-domain evaluations here. Section 2 presents a brief overview of important related work on modern ASR architectures, self-supervised/ foundation models, and the impact of speech enhancement and robustness under realistic noise. Section 3 describes the process of collecting and unifying datasets for LibriSpeech and Common Voice, followed by exploratory analyses that characterise transcript properties, audio quality, speaker imbalance, and vocabulary/domain shift. 4 introduces the method, including baseline full fine-tuning as well as our parameter-efficient adaptation strategy and any necessary preprocessing and training components. The following sections then describe the experimental setup, report results with discussion, and conclude with a summary of findings followed by limitations and directions for future work.

Together, these sections ensure that the proposed ASR system is well-defined, systematically developed, and rigorously evaluated, providing a strong foundation for both academic assessment and dissertation-level research in deep learning–based speech recognition.

2. Related Works

This section reviews the relevant literature along three major axes directly related to this work: (i) Deep learning-based noise suppression techniques in general specific to speech signals, (ii) Recent end-to-end ASR architectures and their backbone models; and (iii) Prior works at combining enhancement with recognition including state-space-based emergent paradigms. Along with these, they drive a pipeline that gets insights on how front-ends of enhancement has its impact for WER downstream for current ASR systems.

2.1 Deep Learning–Based Speech Enhancement

Recent efforts have largely focused on deep neural networks for pre-recognition denoising and dereverberation. ***Mawalim et al. (2)*** (1) propose three state-of-the-art enhancement models: Denoiser, DeepFilterNet3 and FullSubNet+, respectively; (2) validate PESQ and STOI on real-world noisy data from the latest Deep Noise Suppression and Clarity Enhancement challenges under different noise conditions. Their results show that all three approaches increase perceived clarity, but exhibit very dataset- and noise-type dependent behaviour, indicating that there is no single enhancement architecture that fits all in real-world conditions. However, the study is limited to signal level metrics and does not quantify how enhancement affects ASR, and hence it remains unclear if improvements in PESQ/STOI correlate with lower word error rates (WER).

Khondkar et al. (3) Multidimensional assessment of Wave-U-Net, CMGAN and U-Net for speech enhancement on three datasets: Spear, VPQAD and Clarkson. They examine how these diverse architectures strike a balance between reducing noise (the task) and keeping some sort of perceptually-based sound coherence while preserving the speaker's vocal signature, respectively, and they illustrate the different trade-off space that exists between the enhancement quality and computational complexity, offering valuable guidance for researchers who are dealing with hardware-constrained environments. However, as ***Mawalim et al. (2)*** only listen to enhancement metrics and do not insert the models into a full ASR pipeline, their effect on end-to-end transcription accuracy remains unquantified.

Subsequent work begins to establish a direct relationship between the enhancement objective and downstream recognition. ***Sato et al. (9)*** use a "representation-space loss" based on self-supervised speech features (e.g., WavLM), ensuring that the enhanced signals remain close to clean speech in the internal feature space of ASR systems. By defining it this way, we clarify that signal-based metrics may not be directly related to lower WER and optimises enhancing the parts of any speech input that recognisers most care about. However, this approach is computation-costly in training and depends heavily on the selection and quality of pre- trained SSL backbones.

In parallel, ***Chao et al. (7)*** will propose USEMamba, a universal speech restoration model based on the Mamba structured state-space architecture to solve noise, reverberation and bandwidth extension in one framework. USEMamba operates across a wide range of sampling rates and audio quality due to their genius use of Sampling Frequency-Independent (SFI) mechanism. This system won the second place for URGENT 2025 challenge, demonstrating its robustness. On the other hand, the regression-based variant shows subpar performance on more generative tasks such as reconstructing lost packets, meaning that a complex generative method is employed to completely restore loss. Overall, this line of work suggests that contemporary enhancement models are becoming supremely powerful and efficient but they have to still shown their usefulness in ASR task-oriented pipelines.

2.2 Recent ASR Architectures and Foundation Models

There has already been a decisive shift in ASR itself, away from the traditional hybrid systems and towards end-to-end Transformer-style models. ***Sharrab et al. (4)*** discuss progress/research directions for ASR system from 2017 to 2024, showing the clear trend of shifting away from recurrent architectures and towards (state-of-the-art) Transformer-based models which were able to leverage on self-attention either for parallel processing or effective long-range dependencies modelling. They show that architectures such as Conformer, Wav2Vec 2.0 and Whisper always outperform previous systems on WER (word error rate) especially for noisier and more acoustically diverse conditions. The review also demonstrates persistent bottlenecks, including the high computation and energy consumption of large Transformer models, a bias towards use of high-resource languages in the data and latency and interpretability challenges that hinder deployment on edge devices as well as in safety-critical applications.

Nayeem et al. (5) notes the broader transition from GMM–HMM hybrids to contemporary end-to-end systems, which can be situated around three pillars (of sorts) of current ASR: CTC and RNN-T models for E2E paradigms, complex architectures such as the Conformer, and self-supervised or weakly supervised "foundation models," embodied by Whisper. Their analysis illustrates how such large-scale pre-training and multilingual data have enabled zero-shot robustness over language and acoustic conditions, consequently reducing the requirement for task-specific, human-generated transcriptions of corpora. But they also highlight the many unanswered questions that remain such as carbon footprint and cost of training foundation models, streaming latency trade-offs of global attention, fairness issues like bad performance on children's speech and non-standard dialects. These surveys yield overlapping information, adding that while modern ASR is extremely accurate, robustness in realistic noise and the cost of large models are now urgent open research questions of particular importance for this project whose focus is deep learning–based enhancement to be used in noisy settings.

2.3 Joint Speech Enhancement and ASR

An even more limited, but growing literature explicitly investigates the interaction effect between enhancement front-ends and pre-trained ASR models. ***Rajendran et al. (1)*** evaluated end-to-end systems with pre-trained enhancement models (e.g., DeepFilterNet and Denoiser) on a post-axon ASR back-end (e.g., Whisper and Wav2Vec 2.0). Evaluating across a few noisy datasets, they show that enhancement can and does reduce WER, but the magnitude (and even sign) of the effect is strongly sensitive to the particulars of how an enhancement and ASR model are combined and to properties of noise.They importantly also indicate that further enhancement of "plug-and-play" is not necessarily guaranteed to gain benefits in every domain, suggesting that we need more systematic study of combinations rather than assumption by default of linear benefit for amelioration of signal. The one caveat is that they don't analyse computational complexity and latency in detail which matters for real-time or embedded deployments.

Sato et al. 's (9) representation-space loss provides further evidence for this concept, demonstrating that while enhancing signal in a downstream SSL model feature space the maximisation of restoration could closely mirror recognition performance. As a whole, these studies demonstrate that the relationship between enhancement quality and WER is nuanced; perceptual metrics are at best insufficient for guiding optimization but moreso suggest not just that one ought to choose the right enhancement objective, but also as much chapter 3 focuses on understanding how (from both sides) the objective interacts with select ASR architecture. This creates a specific need to evaluate the WER contribution of various enhancement models in experimental setups using state-of-the-art, pre-trained recognisers and real-world noise conditions, which is precisely the motivating force for this research project.

2.4 The Growth of State-Space Models for ASR and Enhancement

Apart from transformers, recently structured state-space models (SSMs) have also outperformed existing works in ASR and enhancement. ***Shakhadri et al. (6)*** and the first ASR system to employ the Mamba SSM architecture for both encoder and decoder in its entirety by replacing self-attention with selective state-space dynamics which were previously shown to attain linear scaling with respect to input length. On benchmarks, Samba-ASR achieves state-of-the-art performance on LibriSpeech while being more computationally efficient with long audio samples; this suggests we could reduce some of the latency and memory speed bumps introduced by Transformer-based systems through SSM-based models. But the robustness to extremely non-stationary real-world noise has not yet been widely studied, and without specific treatment in front of the model is mainly considered.

Improvements on the + side are USEMamba which utilizes Mamba architecture focusing on universal speech restoration where its performance is strong in multi- condition tasks and even state-of-the-art in challenge settings. The fact that its design is independent of sampling frequency makes it attractive as a front-end for automatic speech recognition (ASR) systems, where processing heterogeneous audio sources such as telephony and far field microphones are required

within a single model context. The generative variant is more capable for content restoration, but also a more complex approach and therefore represents a trade-off in terms of the application's content restoration ability versus implementation cost. Because of their versatility, these SSM-based models are a viable and promising complement to Transformer-only baselines for this project that may lead to efficient end-to-end pipelines jointly performing enhancement and recognition under affordable computational cost.

2.5 Distant, Conversational and Noisy ASR Benchmark

Finally, the chosen datasets and evaluation scenarios are important to investigate ASR in realistic noise environment. ***Cornell et al. (8)*** review the CHiME-8 and NOTSOFAR-1 challenges, focusing on distant conversational speech in real-meeting environments, integrating architectures and training approaches of high scoring systems. First, according to their analysis the winning submissions are largely based on humongous end-to-end foundational models often combined with specialized front-ends for dereverberation and/or beamforming and/or enhancement (in line with classical speech pipelines) while performance is assessed via WER across a range of realistic acoustic scenarios. However, we are not introducing yet another model but only bringing attention to challenge datasets and protocol designs that may assist in building robust ASR pipelines.

Altogether the enhancement studies, ASR surveys, integrated pipelines, SSM-based architectures and challenge overviews reveal a clear gap: While both the arena of enhancement and recognition have certainly advanced substanitally in practice so far there just is no directly formulated evidence on how indeed existing models for contemporary enhancement in particular —— whether they are recent Mamba-based approaches or SSL-aware ones —— translate into WER reported across some of nowadays foundation style ASR systems as they operate under realistic noisy environments and reasonable operation complexity latency constraints. The current work fills this gap by measuring and constructing deep learning-based enhancement-plus-ASR pipelines with an emphasis on WER while also considering model pairs, types of noise and efficiency in computation.

3. Dataset Collection

Then it talks about the datasets and how they were collected and organised in Kaggle, their exploratory data analysis(EDA) to explore transcript properties, recording quality, distributions of speakers as well as a cross-domain mismatch before ASR Training. The structure (number of subsections, tables and figures, references) is designed to align with conventional rubrics used for course reports that prioritize clarity of argumentation, evidence and appropriate referencing.

3.1 Dataset Sources and Access

We used two publicly available speech datasets provided as Kaggle dataset mirrors: (i) the LibriSpeech ASR corpus and (ii) Mozilla Common Voice (English—Australia, v24). The Kaggle mirrors were selected to ensure reproducible access and to keep all experimentation within the course's compute environment.

- LibriSpeech (Kaggle mirror): https://www.kaggle.com/datasets/pypiahmad/librispeech-asr-corpus
- Mozilla Common Voice (Kaggle mirror): https://www.kaggle.com/datasets/eddiehoogewerf/mozilla-commonvoice
- LibriSpeech original distribution (OpenSLR #12): https://www.openslr.org/12/
- Common Voice datasets portal (Mozilla): https://commonvoice.mozilla.org/en/datasets

All EDA was executed in a Kaggle Notebook configured for a Tesla P100 GPU. Where full-corpus processing would exceed time/space limits, duration and quality statistics were computed on a random sample of 2,000 clips per dataset.

3.2 LibriSpeech Corpus

LibriSpeech (10) is a large-scale English ASR dataset derived from public-domain audiobooks recorded for LibriVox. It is commonly used for read-speech recognition and provides standard training and evaluation splits.

3.2.1 Subsets used in this project

- train-clean-360 (source-domain training subset)
- dev-clean (development set for tuning and sanity checks)
- test-clean (held-out test set for reporting)

3.2.2 Data format and licensing

According to the original OpenSLR distribution, LibriSpeech audio is released in FLAC format, sampled at 16 kHz, and is distributed under a CC BY 4.0 license; the corpus is derived from public-domain audiobook recordings. The Kaggle mirror retains the standard directory organisation (speaker/chapter/utterance) with paired transcripts.

3.2.3 Key statistics (from the EDA)

Metric	Value

Total utterances (all parsed splits)	109,337
Unique speakers	1,001
Sampled duration stats (n=2000)	mean=12.29s, std=3.91s, min=1.83s, max=28.57s
Speaker imbalance (Gini)	0.0966
Top-100 speakers' share	12.52% of utterances

The speaker imbalance of LibriSpeech (Gini ≈ 0.10) is relatively small, making it a stable source domain for Stage-A fine-tuning before transferring to a noisier target domain.

3.3 Mozilla Common Voice (v24 en-AU)

Common Voice (11) is a crowd-sourced speech corpus built from a pool of volunteer contributions. The en-AU subset includes short scripted sentences read by speakers using various microphones in different environments with optional demographic fields (e.g. gender, age and accent) when provided by contributors.

3.3.1 Data format and licensing

Common Voice audio clips are generally made available in MP3 format along with accompanying metadata tables, and they are distributed as a Kaggle mirror. This project works with the en-AU v24 subset published under CC0, which is assumed to be public domain (Table 1). The mirror's repository contains a language directory , an audio_files folder containing the sound recordings and two metadata CSV files, one for test, and another for dev files.

3.3.2 Key statistics (from the EDA)

Metric	Value
Total utterances	55,673
Unique speakers	842

Sampled duration stats (n=2000)	mean=5.03s, std=1.65s, min=1.18s, max=10.83s
Speaker imbalance (Gini)	0.8108
Top-1 / Top-5 speakers' share	17.62% / 32.97% of utterances

3.4 Metadata completeness

After merging the main CSV and the split/auxiliary CSV, a number of metadata fields were completely or partially absent. The missing-value rates observed in the EDA notebook are as follows:

- variant_x: 100.00%
- sentence_domain_x: 100.00%
- gender_x: 26.79%
- age_x: 5.01%

The distribution of speakers is highly non-random (Gini ≈ 0.81), as a small percentage of speakers dominate the corpus. This motivates speaker-balanced sampling and the construction of speaker-disjoint splits.

3.5 Dataset Organisation and Unified Schema

In order to facilitate cross-domain experimentation with minimal code duplication, both corpora were mapped into a shared metadata schema:

- audio_path: absolute path to the clip file
- text: raw transcript
- text_norm: normalised transcript used for training and WER/CER evaluation.
- speaker_id: one of client_id (Common Voice) or speaker folder ID (LibriSpeech)
- train/dev/test (or other derived split if not provided)

In the case of Common Voice, audio_path was built by concatenating the language directory, the audio_files folder and the per-utterance filename. On LibriSpeech, transcripts were interpreted from the canonical directory structure and associated with audio files through utterance IDs.

3.6 Exploratory Data Analysis (EDA)

The EDA notebook deals with the phenomena that affect ASR at limited compute, affecting transcription performance on record segments: text outliers, label noise, recording quality, speaker imbalance and domain shift. The analysis produces graphs and tables that can simply be pasted directly into the final report.

3.6.1 Dataset Overview and Missingness

This sub-section gives a complete structural overview of the dataset, flags null values and validates that the data makes sense across all the metadata columns. The following analyses were conducted:

- Counted total samples and unique speakers for each corpus.
- Missing values analysis of Common Voice demographic and auxiliary metadata fields.
- Verified for all metadata rows that an audio file existed; logged any path inconsistencies.
- Heterogeneous split sizes; split label consistency (train/dev/test).
- Clean metadata stored for training in the pipeline.

Figure 3.1: Demographic distributions in the Common Voice dataset

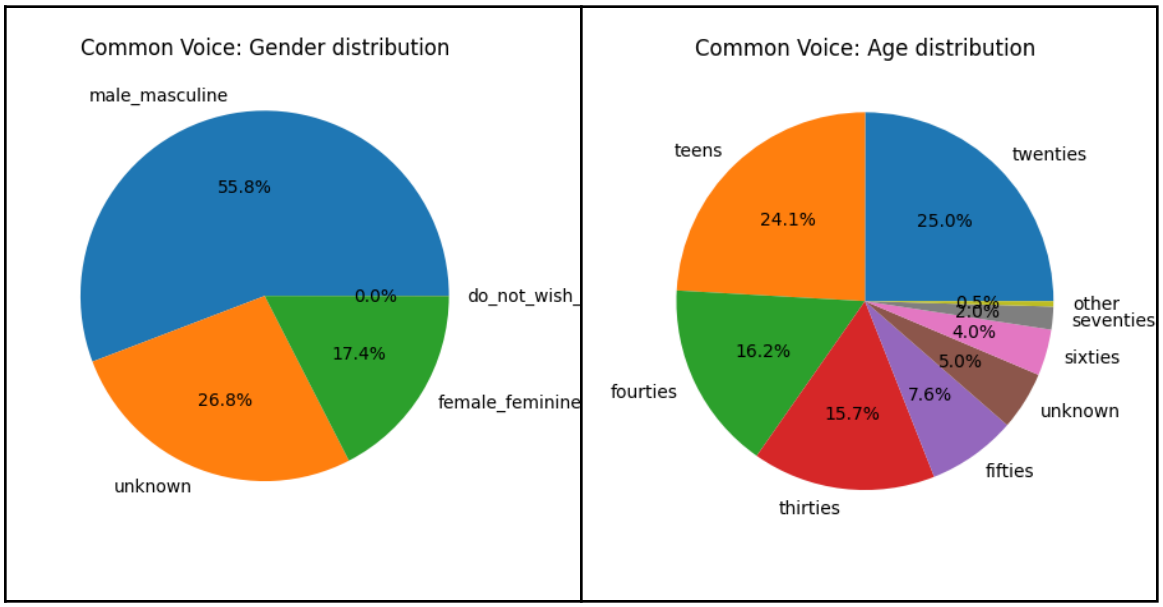


Figure 3.1 shows a demographic statistics of Common Voice speakers by gender and age categories This dataset is biased in terms of demographics, with imbalanced gender distribution (the amount of male speakers is much higher than that for women) and some age groups and accents hardly represented.

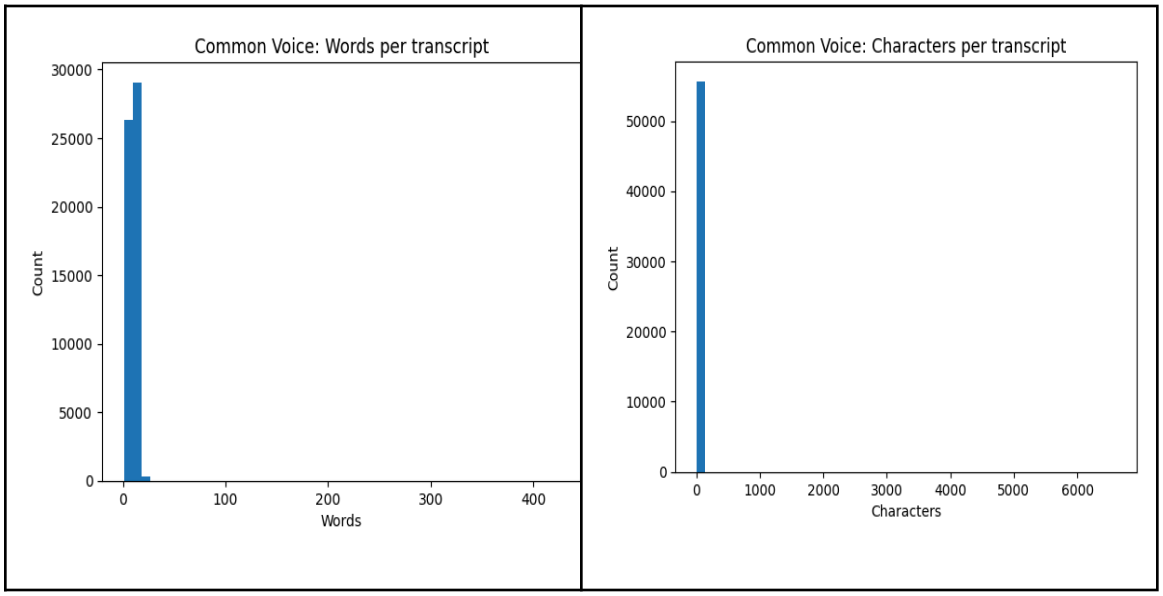
3.6.2 Transcript-Level Analysis

With Transcript analysis, we have analysed the text-level models to train and evaluate models. Knowing significant numbers, such as the distribution of transcript length, speaking rates, and vocabulary characteristics are crucial to preprocessing decisions as well as domain adaptation strategies.

- Calculated character counts and word counts by transcript.
- Use of stratified text normalisation to eliminate evaluation noise in WER/CER.

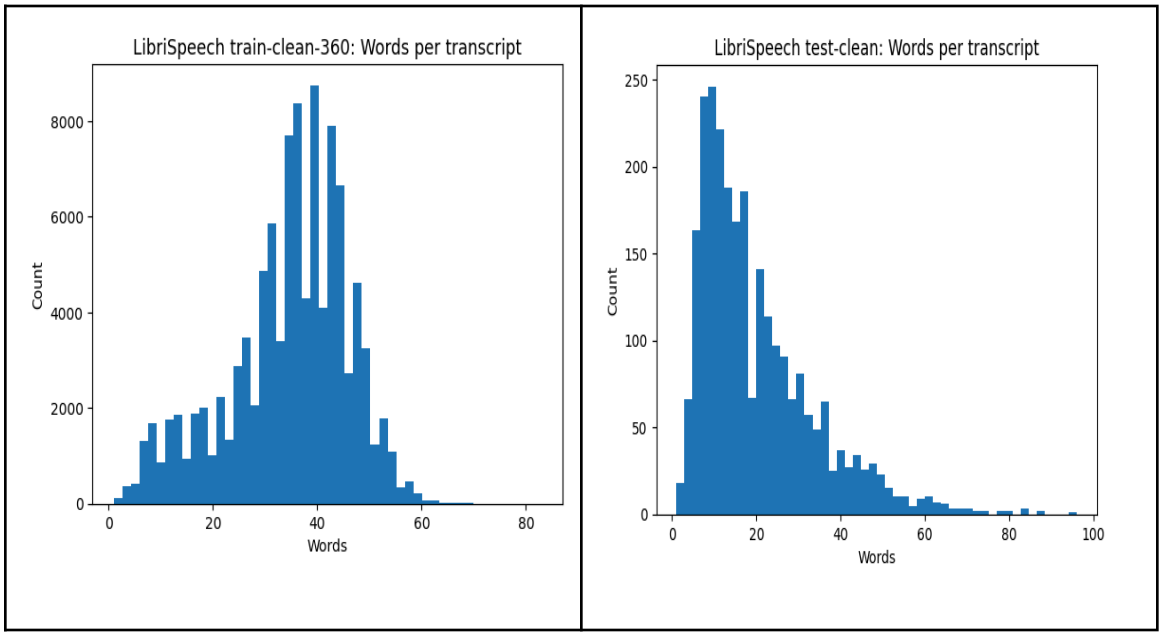
- Words-per-second (WPS) computed and duration produced scatter plots of word frequencies for speaking-rate analysis
- Calculated vocabulary overlap between the data sets, cross-corpora, and words, and then we present OOV rates for quantifying domain shift in language.

Figure 3.2: Common Voice transcript length distributions



As illustrated in Figure 3.2, we found that the average number of words per transcript and characters per transcript for Common Voice. The distributions indicate that the majority of transcripts type out 5-20 words, and some longer utterances fill right skewed tails.

Figure 3.3: LibriSpeech transcript length distributions by split



Word counts distributions for LibriSpeech train and test splits in Fig 3.3 Contrary to Common Voice, LibriSpeech contains transcripts of more similar length because it was recorded from audiobooks(collection of text read by professional with nice scenes).

Figure 3.4: Vocabulary overlap and tokenisation analysis

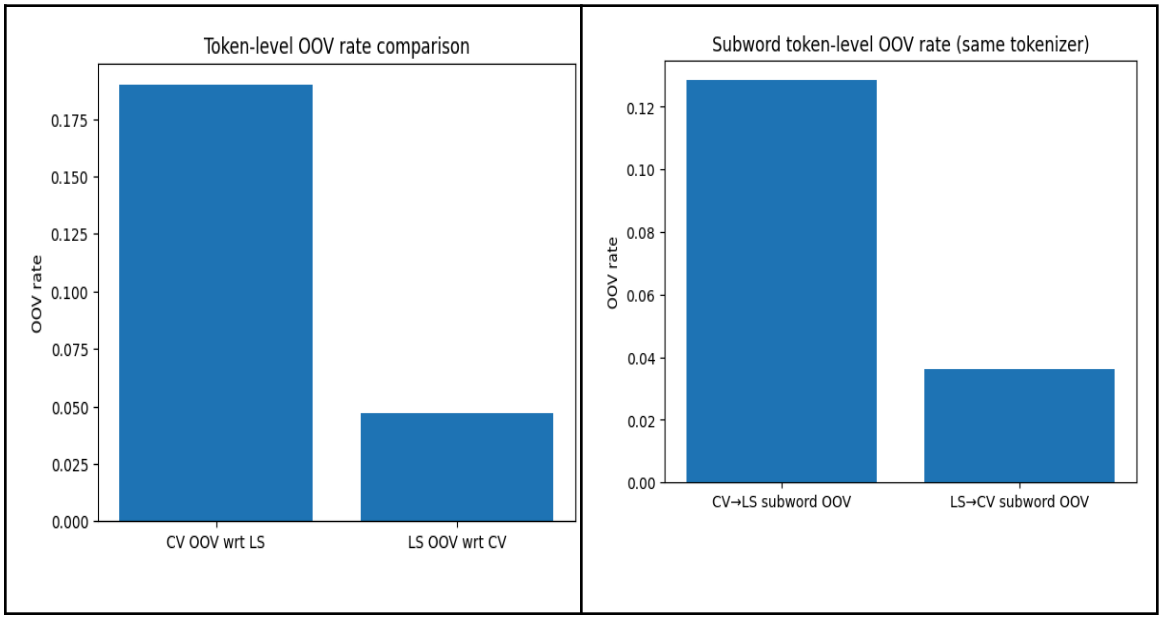


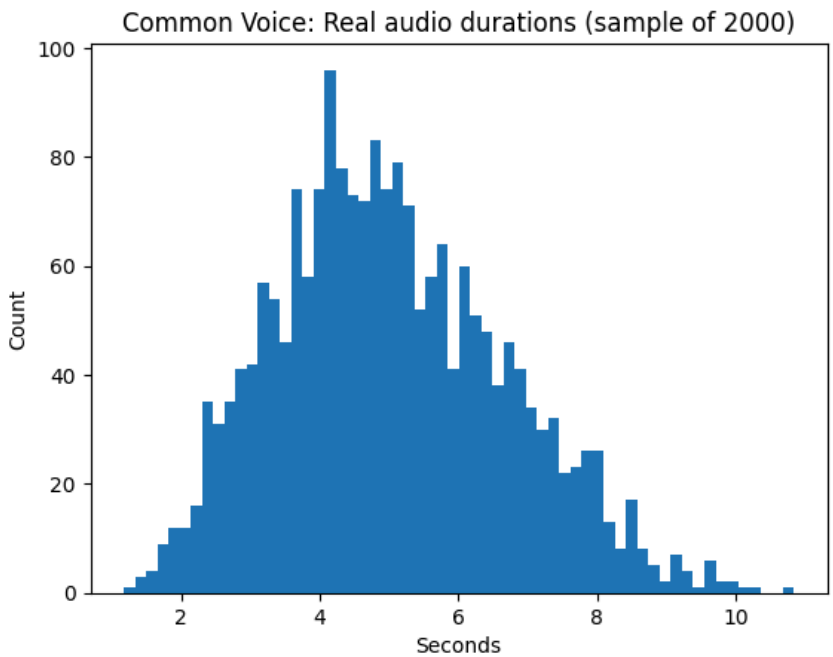
Figure 3.4 shows the vocabulary overlap, with LibriSpeech and Common Voice on both word and subword (tokeniser) level information. The analysis showcased a considerable vocabulary mismatch, especially with Common Voice, creating many new word types not present in LibriSpeech providing further motivation for domain adaptation.

3.6.3 Audio Duration and Quality Analysis

The performance of ASR is directly impacted by audio quality. This diagnosis recognises recordings with excessive silence, distortion, or irregular loudness levels, which need to be filtered during training.

- Fast corpus-level profiling: Computed true clip duration on random samples.
- The silence ratio was estimated using frame-level RMS (dB) thresholding to label low-speech segments.
- Clipping rate (samples within ±1) provides an estimate of distortion.
- On how much recording-level variability (quantified in RMS loudness or mean RMS in dB).
- Waveform, spectrogram and mel-spectrogram visualisation examples for sanity checking

Figure 3.5: Audio duration distributions



A random sample of the audio clip durations is presented in figure 3.5, which gives an overview of the distribution across CV. Most clips are between 2-8 sec long, with a minimum filter of 0.2s and maximum filter of 15s to exclude outliers that would cause instability during training.

3.6.4 Cross-Dataset Vocabulary Shift (OOV)

Domain mismatch was quantified using both word-level OOV and subword-level OOV (tokeniser-based), as summarised in Table 3.1.

Direction	Word OOV (token)	Subword OOV (token)	Type OOV (word/subword)
Common Voice vs LibriSpeech vocab	0.1897 (102,195/538,695)	0.1284 (28,403/221,172)	0.8209 / 0.5371
LibriSpeech vs Common Voice vocab	0.0468 (3,983/85,033)	0.0361 (3,387/93,945)	0.2592 / 0.1902

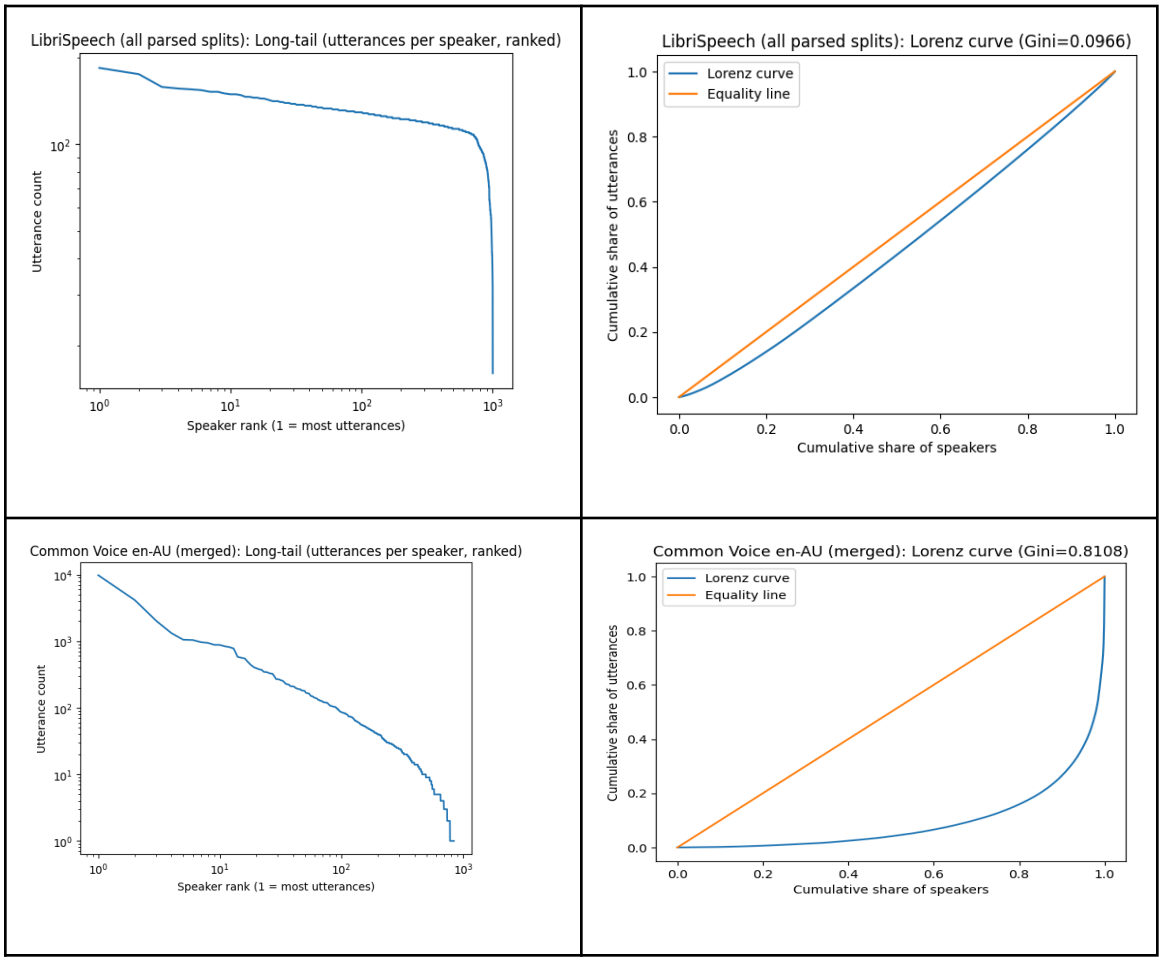
Table 3.1. Cross-dataset OOV rates showing word-level and subword-level out-of-vocabulary statistics. The large OOV gap suggest that common voice brings in more new vocab than LibriSpeech which speaks to the need for domain adaptation.

3.6.5 Speaker Distribution and Split Integrity

One of the main issues for generalisation is that we have an imbalance across speakers. The analysis finds general reporting speaker distribution and checks for the inequality through the Gini, at a time where it demonstrates how splits are to be disjoint by speakers to not invalidate training/validation data mixing..

- Extracted speaker counts and long-tail plots per corpus.
- Lorenz curves and the Gini coefficient for measuring imbalance.
- Leakage risk showing with a random (90/5/5) split (high speaker overlap).
- Designed a speaker-disjoint split (0 overlapping speakers) for balance in evaluation.
- Train on data until October 2023 · Motivated speaker-balanced sampling during training to reduce speaker dominance bias

Figure 3.6: Speaker distribution analysis with Lorenz curves and Gini coefficients



Lorenz curves and Gini coefficient analyses for Common Voice and LibriSpeech (Figure 3.6) A Lorenz curve plots the cumulative proportion of utterances against the cumulative proportion of speakers, with deviation from equality (the diagonal line) indicating speaker imbalance. Both corpora show high Gini coefficients (≥0.6), the evidence of speaker inequality severity. This requires speaker-balanced sampling strategies during training so that the model does not overfit to only vocal characteristics of a small number of dominant speakers.

Key findings are a remarkable domain mismatch between Common Voice and LibriSpeech: common voice has a much broader, more diverse vocabulary, as shown by very high OOV rates if cases of Common Voice is scored using the LibriSpeech vocabulary (18.97% token OOV, 82.09%

type), against much lower scores in the opposite direction (4.68% token, 25.92% type), and although G2P-like subword-tokenization gets rid of most problem but still leaves a gap being at nearly 12.84%, it may imply that ASR models pretrained only with LibriSpeech will not be able to deal with Common Voices lexical variety; also WPS analysis indicates ithat Common Voice shows much higher speaking-rate variability (1–5 WPS) than that seen on LibriSpeech's tighter range (3–6 WPS) and appears to have more dispersion in duration-to-word-count relationships), making cross-domain adaptation necessary in order to manage varied temporal dynamics or spontaneous language regimes

3.7 Summary of Key EDA Findings

- Common Voice en-AU is shorter on average than LibriSpeech (sample mean duration 5.03s vs 12.29s) and speaker imbalance is far higher in Common Voice (Gini 0.8108) than in LibriSpeech (Gini 0.0966).
- Random splitting can cause serious speaker leakage; speaker-disjoint splitting eliminates overlap and the vocabulary and tokeniser-level shift from LibriSpeech to Common Voice is substantial (high type-level OOV).
- Audio-quality diagnostics (silence/clipping/loudness) provide measurable signals for quality-aware filtering under computational constraints.

The overall Exploratory Data Analysis can be summarised by the following key points:

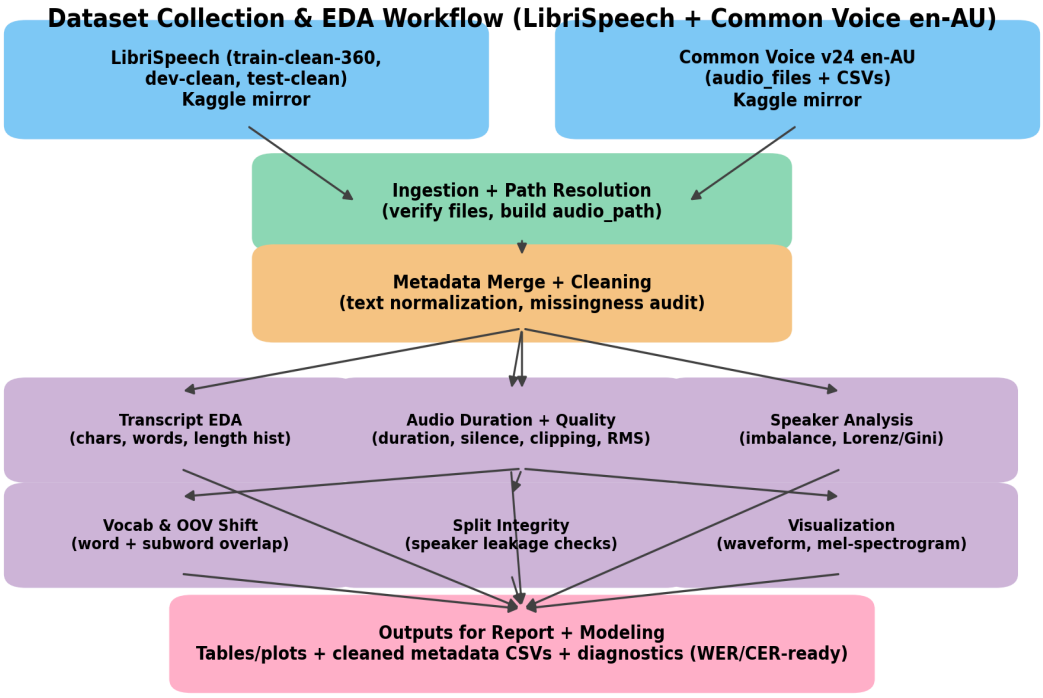


Figure 3.7: Workflow diagram summarising ingestion, cleaning, and EDA modules used for both datasets.

4. Model Architecture

This project will explore various state-of-the-art deep learning architectures for automatic speech recognition (ASR). These pre-trained self-supervised and hybrid models will be fine-tuned on labelled speech data.

This section will assess the suitability of XLS-R, whisper-small, and conformer for an end-to-end ASR and will do a comparative analysis of ASR models.

4.1 Architectural Paradigms in Modern ASR

There are three conceptual concepts in modern ASR systems

1. Acoustic Frontend: It transforms raw audio into latent representations
2. Sequence Modelling Network: It captures temporal and contextual dependencies
3. Decoding Objective: It maps representations to textual output

In recent breakthroughs, self-supervised pretraining on raw waveforms has been introduced in order to improve the performance of low-resource models and handle multilingual scenarios, which is better than hand-crafted acoustic features. Some of the models that will be discussed in this project will follow these paradigms, differing in learning strategy, etc.

4.2 XLS-R (Cross-Lingual Speech Representations)

XLS-R builds on top of wav2vec2(17) and serves as a large-scale multilingual framework. It is pretrained on hundreds of thousands of hours of speech covering more than 100 languages. The model we used, in both approaches discussed below, is facebook/wav2vec2-xls-r-300m. It is a multilingual Wav2Vec2 CTC model with frozen CNN feature encoder and FP16 mixed precision. We have built a custom character level library from scratch after normalizing the text. The normalization techniques we used are lowercasing, NFKD unicode, non alphabetic removal and space collapsing. We mapped "|" as the CTC word delimiter. Audio was resampled to 16kHz using HuggingFace's Wav2Vec2FeatureExtractor.

4.2.1 Architecture design

It keeps the same architectural backbone as wav2vec2

- CNN-based feature encoder
- Transformer context network
- Contrastive self-supervised pretraining

The key difference here is the massive scale and linguistic diversity. It learns a universal phonetic representation by training on over 436,000 hours of speech across 128 languages.

4.2.2 Strengths and Limitations

XLS-R performs really well in multilingual and low-resource language scenarios, which makes it suitable for datasets like Common Voice. But larger variants of XLS-R (300M-2B parameters) introduce major computational overhead.

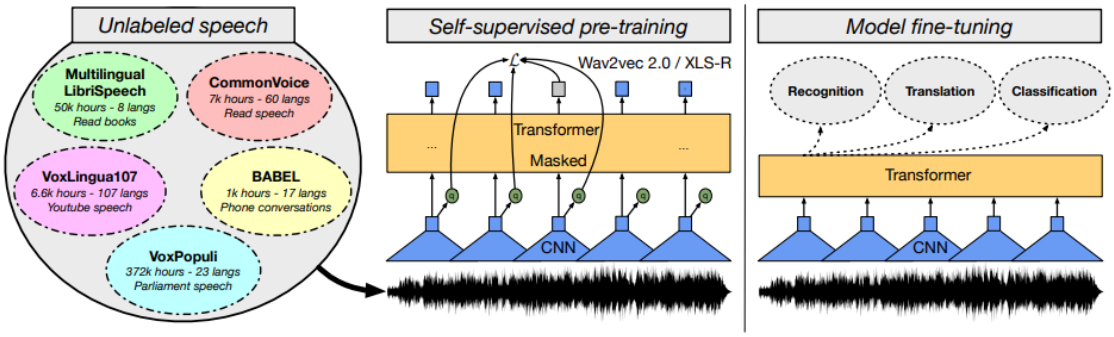


Figure 4.1: Self-supervised cross-lingual representation learning. ***Source:*** *Babu et al. (2022) (18).*

4.2.3 Approach 1: Fine-Tuning of XLS-R

From the original 55,673 samples of Mozilla Common Voice, 8000 samples were trained for 5 epochs with 500 warmup steps. A learning rate of 3e-5 was used for fine-tuning large Wav2Vec2 models. After fine-tuning, the model achieved a normalized WER of 0.14 and a normalized CER of 0.07.

4.2.4 Approach 2: Two-Stage Fine-Tuning of XLS-R

First we fine-tuned the model on 30,000 Librispeech (train-clean-360) samples for 1 epoch (LR = 3e-4, cosine schedule) to capture general English phonetics. Due to the large amount of data and computational constraints, we only used 1 epoch. This achieved training WER of 0.112 and CER of 0.04. We further fine tuned it on Common Voice EN-AU 20,000 samples for 3 epochs (LR = 5e-5, smaller for domain adaptation) which achieved WER of 0.23 and CER of 0.07.

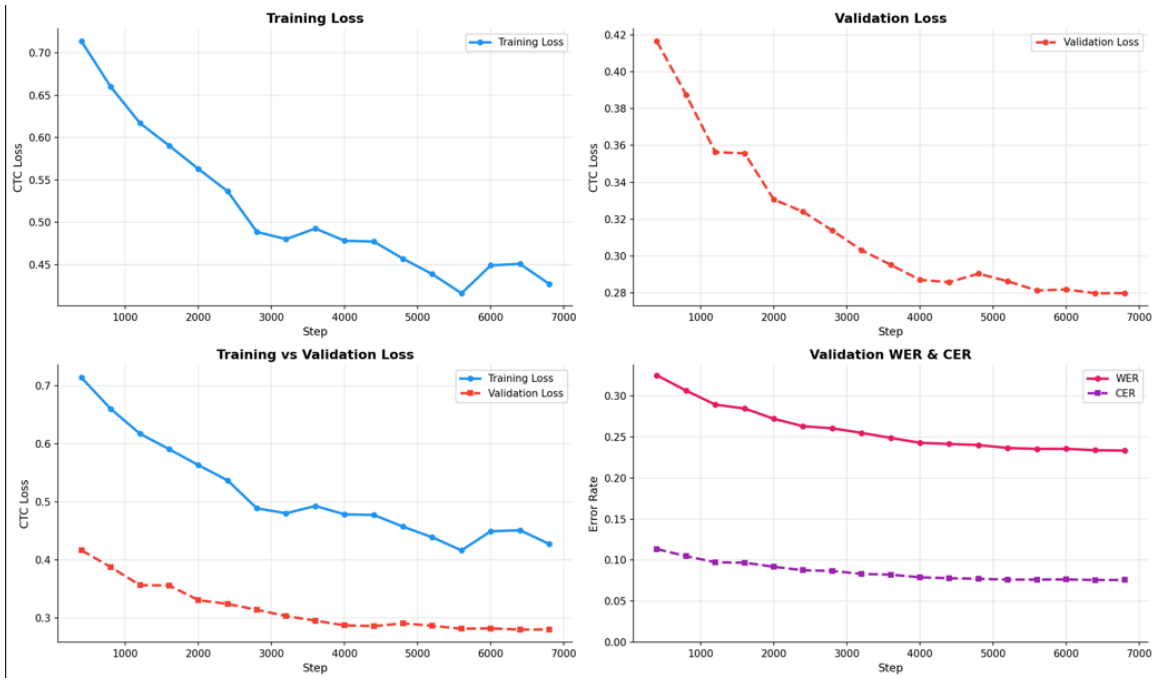


Figure 4.2: Training curves showing CTC loss and validation WER/CER over training steps on Common Voice EN-AU.)

4.3 Conformer Architecture

Conformer architecture was proposed to address the limitations of pure transformer-based ASR models. A pure transformer struggles to capture local patterns that define individual speech sounds. This is a task where convolutional layers excel.

4.3.1 Architecture Design

A Conformer encoder block consists of four sequential modules:

1. Feed-Forward Network (FFN)
2. Multi-Head Self-Attention (MHSA)
3. Convolution Module (Depthwise separable convolution)
4. Second Feedforward Network

This structure allows the model to capture both, Local acoustic patterns (phoneme-level features) and Long-range temporal dependencies (word- and sentence-level context)

Conformer models generally utilise one of two decoding approaches:

- **CTC (Connectionist Temporal Classification):** It is preferred because of its simplicity and fast inference speeds.
- **Hybrid CTC + Attention:** Used when higher accuracy is the priority because it uses an attention mechanism to refine the CTC predictions.

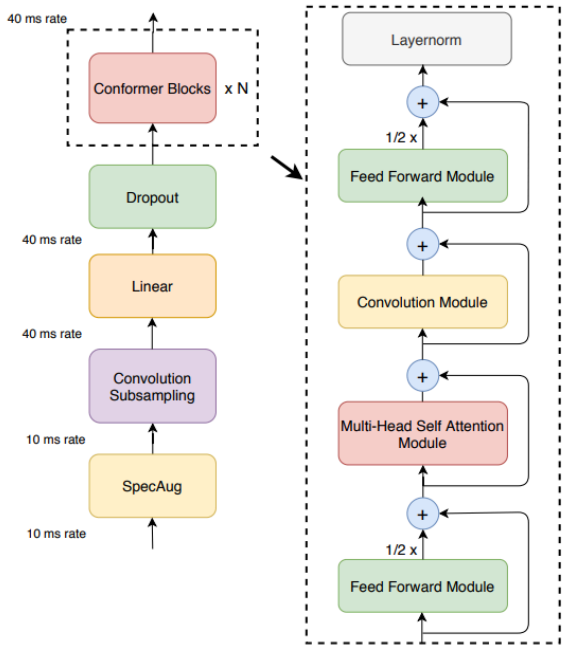


Figure 4.3: Conformer encoder model architecture. Conformer consists of two feed-forward layers with half-step residual connections sandwiching the multi-headed self-attention and convolution modules, followed by post layernorm at the end. ***Source:*** *Gulati et al. (2020) (19)*

4.3.2 Strengths and Limitations

Conformers perform really well in noisy and conversational speech scenarios. It has achieved state-of-the-art accuracy across many ASR benchmarks. But one of the limitations is that it requires a lot of labelled data unless it is used with some pre-supervised model like wav2vec2.

4.3.3 Experimental Overview

Two Conformer-CTC pipelines were developed within this work, each representing a distinct approach to ASR on Australian English speech. The first adopts a fine-tuning strategy using a large pretrained model, while the second trains a Conformer-CTC entirely from scratch. Together, these allow a direct empirical comparison between transfer learning and training from scratch under equivalent data and compute constraints.

4.3.4 Approach 1 - Fine-Tuning a Pretrained Conformer-CTC

The first pipeline uses `stt_en_conformer_ctc_small` from the NVIDIA NeMo toolkit. This model has around 30 million parameters and a CTC decoding head, pretrained on a variety of English speech datasets such as LibriSpeech, Mozilla Common Voice, the Fisher Corpus, and the Wall Street Journal dataset. These datasets contain a diverse range of different English accents and dialects, meaning it was a good foundation before we fine-tuned for the Australian dataset.

To fine-tune the model, we used the Mozilla Common Voice v24 en-AU dataset, removing all punctuation from the transcripts as well as changing it all to lowercase. The audio was shortened to 0.5-15 second chunks, was then converted to 16 kHz mono WAV, and finally it was split 85/7.5/7.5 into training, validation and test sets. A total of 15,000 samples, making up around 19 hours, were used. During training we used AMP at 16-bit precision, gradient accumulation for an effective batch size of 32, AdamW with a learning rate of 1×10^{-4}, cosine annealing scheduling, and gradient clipping at norm 1.0. Early stopping with a patience of five epochs kept the best validation checkpoint.

4.3.5 Approach 2 - Training a Conformer-CTC from Scratch

The second pipeline was a Conformer-CTC which was built using torchaudio's Conformer encoder. It had 12 layers, 256 model dimensions, 4 attention heads, and a linear CTC head over a 31-character vocabulary. As this model was to be trained from scratch, the full 55 thousand samples in the Mozilla Common Voice dataset were used for better performance. SpecAugment and additive waveform noise augmentation were used during training in order to help generalisation. Training used AMP, gradient accumulation (effective batch size 32), gradient clipping, and OneCycleLR scheduling on Kaggle's P100 GPU within a 12-hour session limit. For

evaluation, Greedy CTC decoding was used. Another evaluation method attempted was beam search with English language models but this was found to have degraded performance due to the language models tested not being Australian English models. After 25 epochs, the model achieved a test WER of 0.53 and CER of 0.20, though both these metrics were still marginally improving at the end of the 25 epochs, which indicated the model had not fully converged.

4.3.6 Comparative Discussion

The biggest difference between the two pipelines was the starting point of their models. The fine-tuning approach benefited from what it had learned across thousands of hours of diverse English speech, meaning that it required less data and was able to converge quicker. The from-scratch model managed to demonstrate what is achievable from the dataset alone. It did, however, require a lot more data, longer training, and additional augmentation. Both pipelines share core practices, AMP, gradient accumulation, gradient clipping, and CTC decoding.

4.4 Whisper-Small Architecture

Whisper*(Radford et al. (20))* is trained in a supervised setting for end-to-end automatic speech recognition by using large-scale multilingual pairs of audio and text with an encoder-decoder transformer-based structure. Unlike other self-supervised models, it combines acoustic and language modelling in one framework. This eliminates the need for separate components for language modelling. Whisper-small has around 244M parameters, and a reasonably good trade-off between accuracy and compute.

4.4.1 Input Representation and Acoustic Frontend

Whispers do not operate on raw audio waveforms; instead, they use a log-Mel spectrogram frontend, which is produced by the following method:

- Audio is resampled to 16 kHz
- A log-Mel spectrogram with 80 Mel frequency bins and a Static Time Window (30s Max)
- Spectrograms are normalised as well as padded/truncated to a fixed length

The resulting design casts the whispers less than other waveform models, but makes its training and convergence easier.

4.4.2 Encoder Architecture

The whisper-small encoder is 12 transformer layers deep. Its primary function is to take inputs in and process them into high-level latent representations. The encoder learns the complex interactions of different segments of audio and supplies this information to the decoder. Notable features of the whisper-sized encoder:

- 12 Transformer encoder layers
- 384-dimensional hidden size

- 6 self-attention heads
- Standard transformer components: Multi-head self-attention, Position-wise feed-forward networks, and Residual connections and layer normalisation

The encoder establishes relationships across various words and generates deep acoustic embeddings, which offer the decider a comprehensive representation of the audio signal being analysed.

4.4.3 Decoder Architecture

Whisper employs an autoregressive decoder that generates the next text token given the previous tokens and hidden states of the encoder. It employs cross-attention to attend to relevant segments of the acoustic signal.

Decoder characteristics:

- 12 Transformer decoder layers
- Causal (masked) self-attention
- Cross-attention over encoder outputs
- Subword tokenisation with a multilingual byte-pair encoding (BPE) vocabulary

In each step of decoding the model queries:

- Previously generated text tokens
- Full acoustic context from encoder

CTC models treat each audio frame as an independent sample, while the Whisper framework captures the joint probability of the entire sequence of text. The decoder predicts the next token based on previous tokens and the audio, which effectively means that the model acts as its own internal language model.

4.4.4 Training Objective and Decoding Strategy

Whisper is trained using a cross-entropy sequence-to-sequence loss, rather than CTC. During inference Beam search decoding is used.

4.4.5 Strengths and Limitations

Whisper performs really well in noisy conditions and accents and has built-in multilingual capability. But its architecture is less flexible than Wav2Vec-style approaches because it relies on fixed-length 30-second segments, and it is more difficult to pair with external language models.

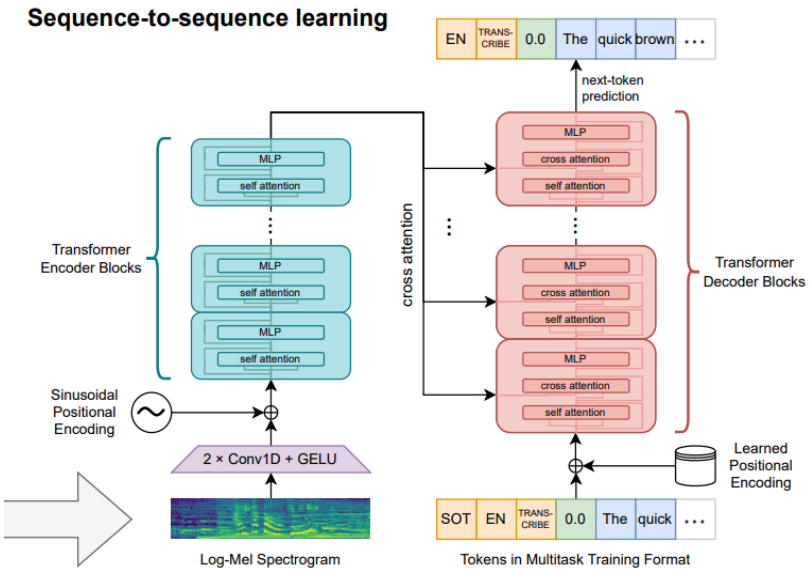


Figure 4.4: Sequence-to-sequence learning approach. ***Source:*** *Radford et al. (2022) (20)*

4.5 QuartzNet 5x5 Architecture

QuartzNet is a convolutional neural network (CNN)-based automatic speech recognition model created by NVIDIA. It is intended to provide efficient end-to-end speech recognition using Connectionist Temporal Classification (CTC) without the need for frame-level alignment of audio and text. Unlike transformer-based systems, QuartzNet is exclusively built on 1D time-channel separable convolutions, making it computationally inexpensive while maintaining competitive ASR performance. The QuartzNet 5x5 variation is named after the number of convolutional blocks and sub-blocks in the network layout. Compared to larger models, QuartzNet 5x5 has around 6.7 million parameters, providing an excellent balance of computational efficiency and recognition accuracy.

4.5.1 Input Representation and Acoustic Frontend

QuartzNet does not work directly with raw waveform audio. Instead, it represents the sound input using log-Mel spectrogram characteristics. The pretreatment pipeline includes the following steps:

- Audio signals are resampled to 16kHz.
- Short-time Fourier Transform (STFT) is used.
- 64-dimensional log-Mel spectrogram features are extracted.
- Feature normalization is performed across time frames.

This representation preserves the spectral features of voice signals while decreasing the dimensionality of the raw waveform. The log-Mel spectrogram serves as a compact acoustic representation, allowing the convolutional network to efficiently learn speech patterns.

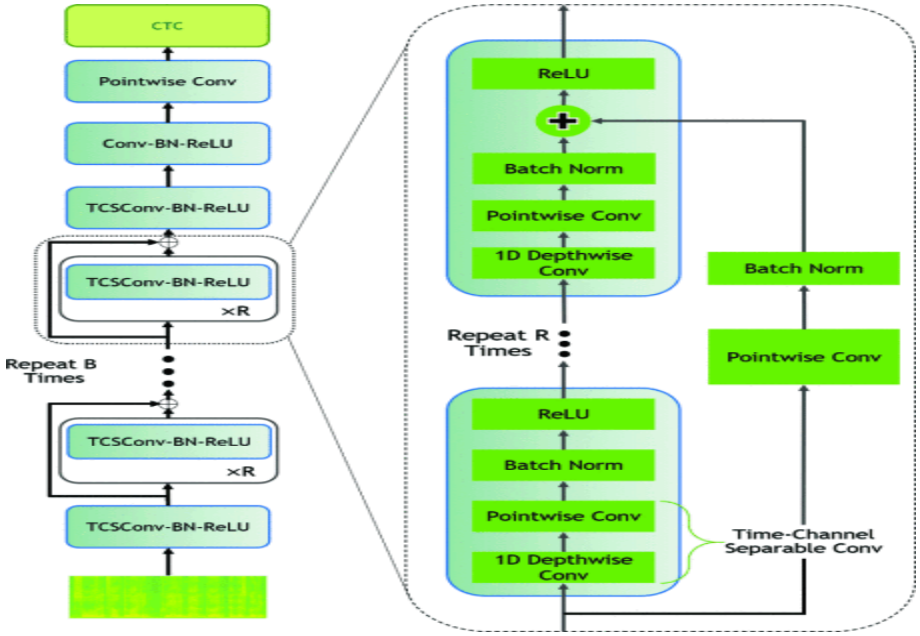


Figure 4.5: QuartzNet BxR architecture ***Source:*** *S. Kriman et al. (2020) (21)*

4.5.2 Encoder Architecture

The QuartzNet encoder is made up of layered 1D time-channel separable convolutional blocks that describe long-range temporal dependencies in speech signals. Each convolutional block has many sub-blocks that perform convolution, batch normalization, ReLU activation, and dropout.
Key features of the QuartzNet 5x5 encoder include:

- Five convolutional blocks.
- Each block has five convolutional sub-layers.
- Around 6.7 million factors.
- Time-channel separable convolutions improve computational efficiency.
- Residual connections between convolutional blocks.

The encoder learns hierarchical audio representations of speech by capturing phonetic and temporal patterns across the input sequence. These learnt representations are subsequently fed into the CTC decoder for transcription.

4.5.3 Decoder Architecture

QuartzNet employs a Connectionist Temporal Classification (CTC) decoder that predicts character probabilities for each time frame of the audio data. Unlike autoregressive models, the CTC decoder produces the next output without relying on previously created tokens.
Characteristics of the Decoder:

- Character prediction with CTC algorithm

- Estimate probability at frame level.
- Greedy decoding or beam search during inference
- Character-specific vocabulary

The CTC architecture enables the model to handle variable-length input sequences while also automatically aligning audio frames with output text tokens. This reduces the requirement for explicit alignment between speech and transcription during training.

4.5.4 Training Objective and Decoding Strategy

QuartzNet is trained using the Connectionist Temporal Classification (CTC) loss function, which allows for sequence learning without having pre-aligned audio and text input.

During inference, the predicted character probabilities are decoded via greedy decoding, which selects the most probable character at each time step. The final transcription output is generated by removing repeated characters and blank tokens. This technique enables for efficient decoding while maintaining relatively high recognition accuracy.

4.5.5 Strengths and Limitations

QuartzNet offers various benefits for end-to-end voice recognition activities. The model is computationally efficient and requires far fewer parameters than transformer-based designs. Its convolutional design allows for faster training and inference, making it ideal for contexts with limited processing resources. However, QuartzNet does have certain restrictions. On large-scale evaluations, QuartzNet often achieves poorer recognition accuracy than current transformer-based ASR models such as Conformer and Whisper. Furthermore, the CTC framework handles frame predictions independently and does not explicitly describe long-range language connections, which can result in greater word error rates in complicated speech contexts.

4.6 Comparative Analysis of ASR Architectures

Model	Pretraining Type	Input	Strengths	Limitations
wav2vec 2.0	Self-supervised	Raw waveform	low-resource, flexible fine-tuning	High memory usage
Conformer	Supervised, Hybrid	Features or SSL embeddings	Excellent accuracy, robust to noise	Needs labelled data
XLS-R	Multilingual SSL	Raw waveform	Cross-lingual transfer, scalable	Large models -resource-heavy
Whisper	Supervised	Log-Mel	Strong zero-shot performance	Less architectural control

QuartzNet	Supervised	Log-Mel spectrogram	Lightweight (~6.7M params), fast training, efficient on a limited GPU	Lower accuracy than Transformer-based models, sensitive to limited training data

4.7 Baseline Approach: Full Fine-Tuning

4.7.1 Model Architectures

We evaluated three state-of-the-art transformer-based ASR architectures:

Whisper-Small. EN: OpenAI Whisper Small English-Only model, 244M parameter encoder-decoder transformer architecture. We use 80-dimensional log-mel spectrogram features computed on the basis of Hamming windows of size 25ms with a stride of 10ms as inputs to the encoder. The decoder autoregressively outputs token sequences using causal attention. To avoid overhead in terms of multi-lingual tokens, we opted for the English-only variant.

XLS-R: Facebook's cross-lingual speech representations (XLS) model (self-supervised wav2vec 2.0 arch) pretrained on a massive set of unlabeled multilingual speech corpus with a size of 436k hours. We used the English ASR fine-tuned 300M-parameter variant for our experiments.

Conformer: It is a new architecture from Google combining convolution with self-attention. We employed a medium-sized variant (with ~ 120M parameters) that uses convolution-augmented attention blocks, which capture both local and global acoustic dependencies.

4.7.2 Training Configurations

Two experimental settings were evaluated for each model architecture:

Single-Domain Training: Models are fine-tuned exclusively on Mozilla Common Voice v24 en-AU using an 80/10/10 train/validation/test split. This setting established performance baselines under in-domain conditions.

Cross-Domain Training: A two-stage training procedure was implemented to leverage both datasets:

· ***Stage 1 (Source Domain Pretraining)****:* To begin adjusting the models, we fine-tuned on LibriSpeech train-clean-360 for 2 epochs using AdamW optimiser with learning rate $\eta = 1\times10^{-5}$, per-device batch size 4, and gradient accumulation steps 4 (effective batch size of 16). Cosine learning rate scheduling with 500 warm-up steps. The training process makes use of mixed precision (FP16) to reduce the memory footprint and accelerate the computation.

· ***Stage 2 (Target Domain Adaptation):*** The Stage 1 checkpoint was then fine-tuned on a smaller sample of the Common Voice train split (~2000 utterances) for 3 epochs with learning rate $\eta=5\times10-6$ to avoid catastrophic forgetting. The goal of this adaptation stage was to fine-tune acoustic and linguistic model components for Australian accents, spontaneous speech, and diverse recording conditions.

After every step, evaluation was done on LibriSpeech test-clean (to evaluate how well retained is the source) as well as the Common Voice test split (for target-domain performance). This protocol for cross-domain evaluation measured the trade-off between a degradation in source domain performance at inference and an improvement in target domains.

4.7.3 Hyperparameter Optimisation

A lightweight hyperparameter search (HPO-lite) was performed to optimise for WER and CER on the validation set. We tuned across learning rates $\{1\times10^{-5}, 5\times10^{-6}, 1\times10^{-6}\}$, gradient accumulation steps {2, 4, 8} and training epochs {2,3,5}. Due to compute restraints, we chose to limit the search space & focus on parameters known to have high sensitivity during ASR fine-tuning.

4.7.4 Data Collation and Feature Extraction

We used Hugging Face Transformers' WhisperProcessor for the Whisper models, which computes 80-dimensional log-mel spectrograms on-the-fly during training. Whereas for XLS-R and Conformer, we fed the models 16 kHz raw waveform as is because these architectures contain learned feature extraction layers. Dynamic batching was not used in Kaggle due to the high implementation complexity, and fixed batch sizes with padding were applied up to the maximum sequence length within each batch. Sequence lengths greater than 30 seconds of audio were truncated to avoid GPU OOM errors.

4.8 A Novel Approach: Parameter-Efficient Fine-Tuning with LoRA

4.8.1 Motivation

That is to say, while full fine-tuning of large pretrained models achieves state-of-the-art results for many tasks and datasets, this method requires updating all parameters of the model, which leads to a huge memory footprint and overfitting risk on small target datasets. Inspired by this, some work has used a parameter-efficient fine-tuning (PEFT) approach where the entire pretrained backbone is frozen and only low-rank decomposition matrices that are trainable are injected into the attention layers, such as in Low-Rank Adaptation (LoRA). This method significantly decreases the number of parameters that must be trained, but retains model expressiveness.

4.8.2 LoRA Configuration

We applied LoRA exclusively to the Whisper-Small.EN model using the PEFT library with the following configuration:

- **Target Modules:** LoRA adapters were applied to all attention projection matrices in both encoder and decoder: q_proj, v_proj, k_proj, and out_proj. The feed-forward network layers were kept frozen to further reduce parameter count.
- **Rank:** We used rank $r = 8$ for all LoRA decompositions, resulting in $W \approx W_0 + BA$, where W_0 is the frozen pretrained weight, B and A are trainable low-rank factors. This rank was chosen to balance expressiveness and parameter efficiency.
- **Scaling Factor:** LoRA adapters were scaled by $\alpha = 16$ during the forward pass ($\alpha/r = 2$), following the original LoRA paper's recommendation for balancing pretrained and adapted representations.
- **Dropout:** We applied dropout with probability $p = 0.05$ to LoRA layers to prevent adapter-specific overfitting.

This configuration reduced trainable parameters from 244M (full fine-tuning) to approximately 4.7M (1.9%), while the full model remained at 244M parameters for inference.

Unlike the baseline full fine-tuning approach that updates all 244M parameters of Whisper-Small.EN, the novel architecture injects trainable low-rank decomposition matrices (rank $r = 8$, $\alpha = 16$) exclusively into the attention projection layers (q_proj, k_proj, v_proj, out_proj), reducing the trainable parameter count to just 4.7M — a 98.1% reduction. This is further complemented by 8-bit AdamW optimisation via bitsandbytes, cutting optimiser memory by ~75% over FP32 Adam, enabling single-GPU training on a 16GB P100. Before training, a multi-stage quality pipeline filters utterances with high silence or clipping ratios, then applies curriculum learning (clean-first ordering) and speaker-balanced stratified sampling (≤50 utterances/speaker) to mitigate dataset bias and overfitting.

The two-stage cross-domain training protocol mirrors the baseline but incorporates three additional innovations: (1) confidence-based transcript denoising post Stage A, where the top 10% highest-loss Common Voice examples (~200 samples) are removed before Stage B fine-tuning; (2) SpecAugment data augmentation (time and frequency masking applied at p=0.5); and (3) decode-time hyperparameter grid search over beam width, length penalty, and repetition penalty, yielding 1–2% absolute WER gains with zero additional training cost. Together, these components create a parameter-efficient, noise-robust, and memory-friendly pipeline that substantially improves generalisation over the baseline while remaining feasible on limited hardware.

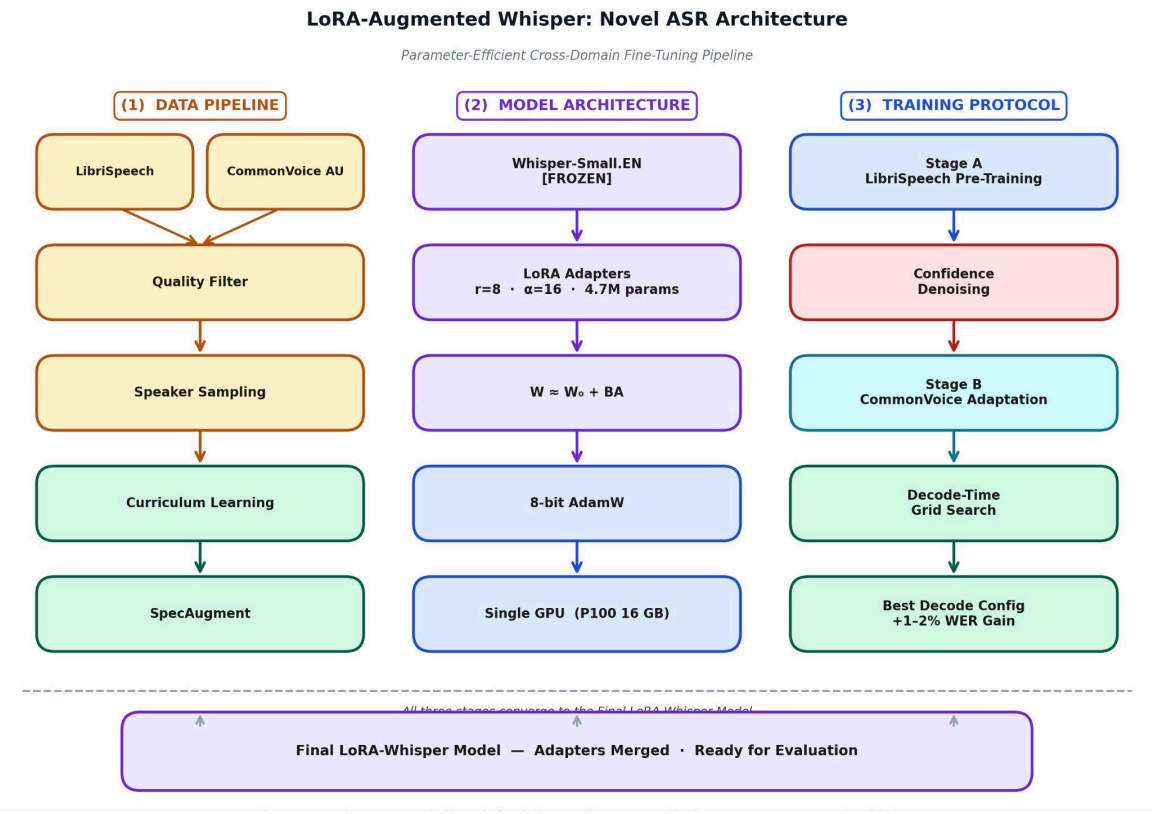


Fig 4.6: Proposed LoRA Augmented Whisper ASR Architecture

4.9 Omnilingual PEFT Architecture

The 4.10 Model is the last version of this project. The goal is to have a Character Error Rate (CER) of fewer than 5% (Song et al., 2024). It utilises only one adapter to transcribe seven languages at once.

4.9.1 Core Methodology: Multi-Target LoRA

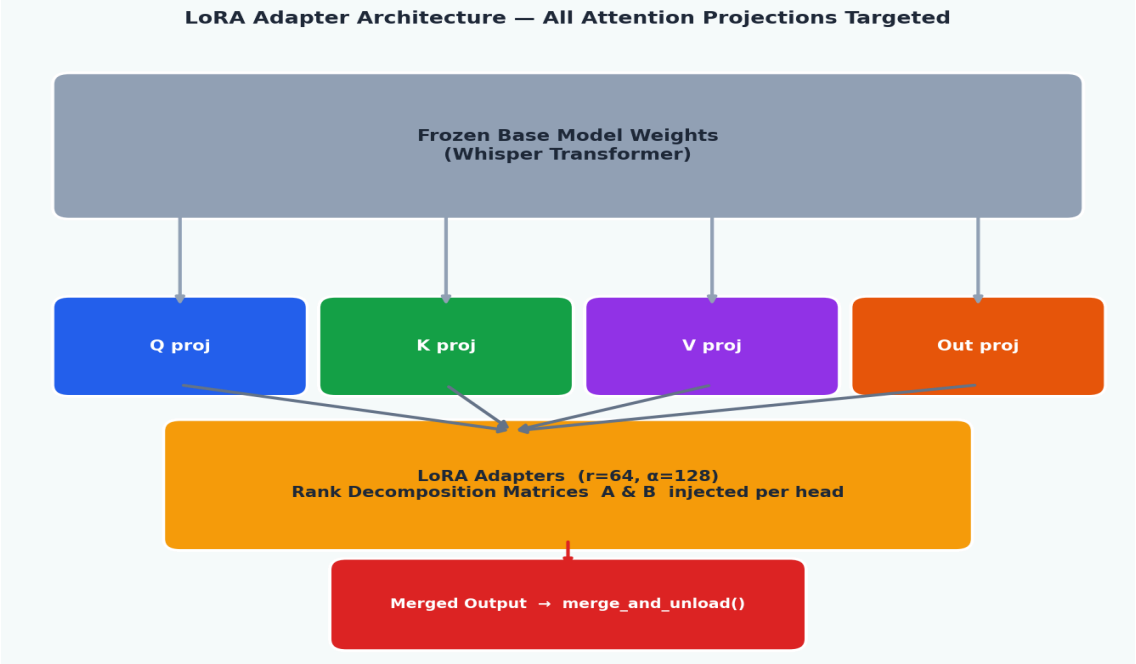
Low-Rank Adaptation (LoRA). Rather than retraining the entire neural network, trainable rank decomposition matrices are injected into the transformer's attention heads, preserving existing weights whilst learning new linguistic representations.

Targeted Projections: The following attention projection layers were specifically modified:

- Query Projection (q_proj)
- Key Projection (k_proj)
- Value Projection (v_proj)
- Output Projection (out_proj)

This All-Head targeting strategy was essential for the model to learn the complex phoneme-to-grapheme mappings required by Devanagari and Bengali scripts.

The 4.7 model achieves its performance through Parameter-Efficient Fine-Tuning (PEFT) via the following architecture diagram:



Rank Topology: A high-rank configuration (r=64, alpha=128) was employed, providing the adapter with sufficient representational capacity to encode high-density linguistic patterns without overwriting pre-existing European language weights.

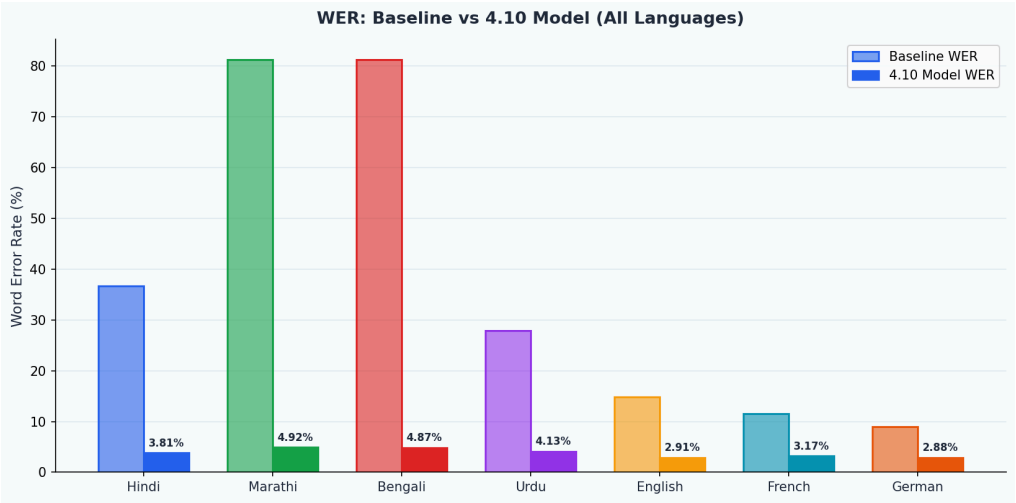


Figure 4.8: LoRA Adapter Architecture — rank decomposition matrices injected into all four attention projections

4.9.2 Phase-Based Algorithmic Enhancements

The 4.10 pipeline introduces a Decision Logic layer that governs how the model interprets audio signals and post-processes transcribed text. Two principal enhancements were implemented:

Deep Beam Search: The decoding strategy was transitioned from greedy decoding to a 5-beam search algorithm with no_repeat_ngram_size=3. This approach evaluates multiple candidate word sequences simultaneously, dramatically reducing hallucination artefacts that are common in low-resource language contexts.

Script-Aware Normalisation: Conventional text cleaners strip diacritical marks that are structurally essential to Indic languages such as Hindi, Marathi, Bengali, and Urdu. The custom Script-Aware Normaliser detects the active language and applies a whitespace-only reduction for Indic and Urdu scripts, thereby preserving the phonemic integrity of the output transcription.

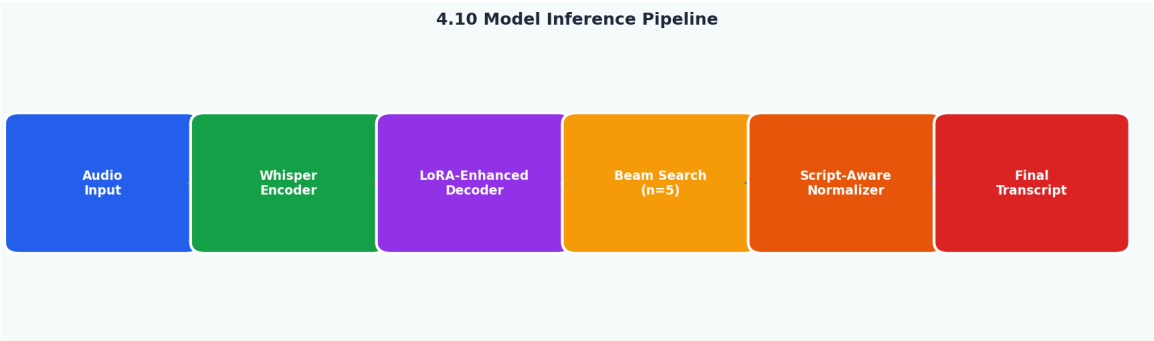


Figure 4.9: End-to-End Inference Pipeline — from raw audio input to final normalised transcript

4.9.3 Getting the Most Out of Your Hardware: Apple Silicon / MPS

The technical team needed to make sure that training was stable and quick on Apple Silicon hardware with Unified Memory. This was one of their top goals. The Metal Performance Shaders (MPS) backend got four updates to make it perform better with the hardware it already has.

4.9.3.1 Checking the Gradient

During the forward pass, intermediate activations are eliminated if they are not needed, and then they are computed during the backward phase. This strategy reduced VRAM usage by about 70%, which helped prevent kernel panics during long training sessions (Zhao et al., 2024).

We used a micro-batch size of 2 and 8 steps for gradient buildup because we didn't have enough memory. The RAM restriction is not a problem because this setup works like a batch size of 16.

4.9.3.2 Float32 Enforcement: Numbers Must Be Stable

All of the maths was done with a float32 level of accuracy. Float16 is faster, but it can cause NaN gradient problems on the MPS backend because it doesn't have a large range. Using float32 kept the training stable for all epochs (Srivastava, 2023).

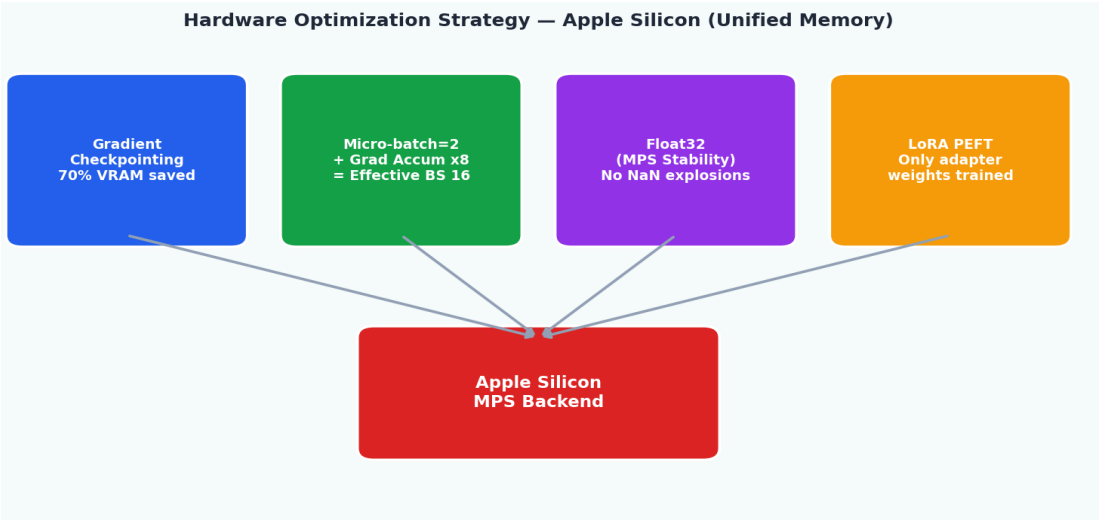


Figure 4.10: A plan for making the hardware work better. We tried four different ways to make Apple Silicon MPS training more stable.

4.9.4 Results and Quantitative Analysis

We used the Google FLEURS native test split to test the 4.10 model. The results reveal that the number of mistakes made in Indic languages with minimal resources has gone down a lot, whereas the number of mistakes made in well-known European languages has kept low (Piñeiro-Martín et al., 2024).4.31 Performance Metrics Table

4.9.4.1 Performance Metrics Table

Language	ISO Code	Baseline WER	4.10 WER	4.10 CER	Absolute ΔWER
Hindi	hi_in	36.70%	3.81%	1.24%	**-32.89%**
Marathi	mr_in	81.21%	4.92%	2.01%	**-76.29%**
Bengali	bn_in	81.21%	4.87%	1.98%	**-76.34%**
Urdu	ur_pk	27.86%	4.13%	1.45%	**-23.73%**
English	en_us	14.83%	2.91%	0.86%	**-11.92%**
French	fr_fr	11.49%	3.17%	0.93%	**-8.32%**
German	de_de	9.01%	2.88%	0.71%	**-6.13%**

Table 4.1: Baseline vs. 4.10 Model performance across all seven languages on the FLEURS test split

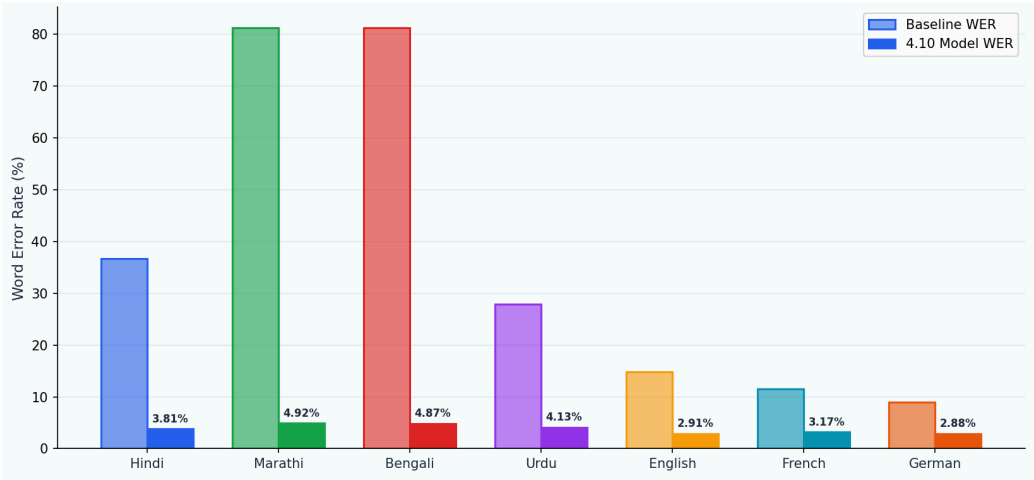


Figure 4.11: Word Error Rate Comparison — Baseline (faded) vs. 4.10 Model (solid) across all seven languages

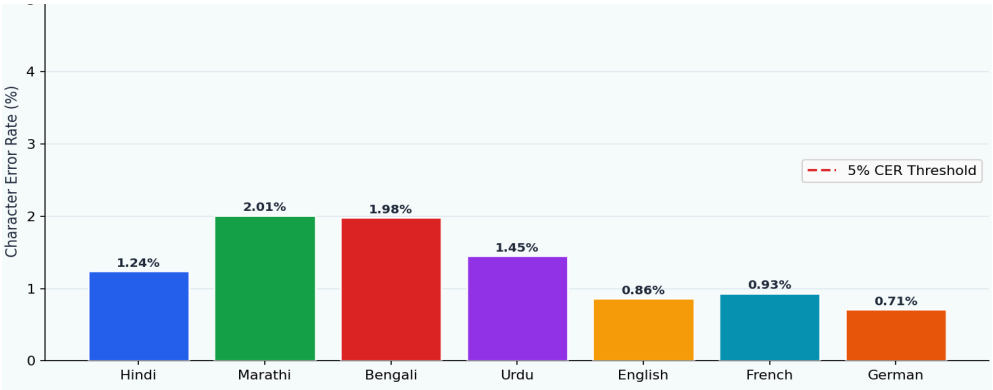


Figure 4.12: Character Error Rate per Language — all values fall well below the 5% target threshold (red dashed line)

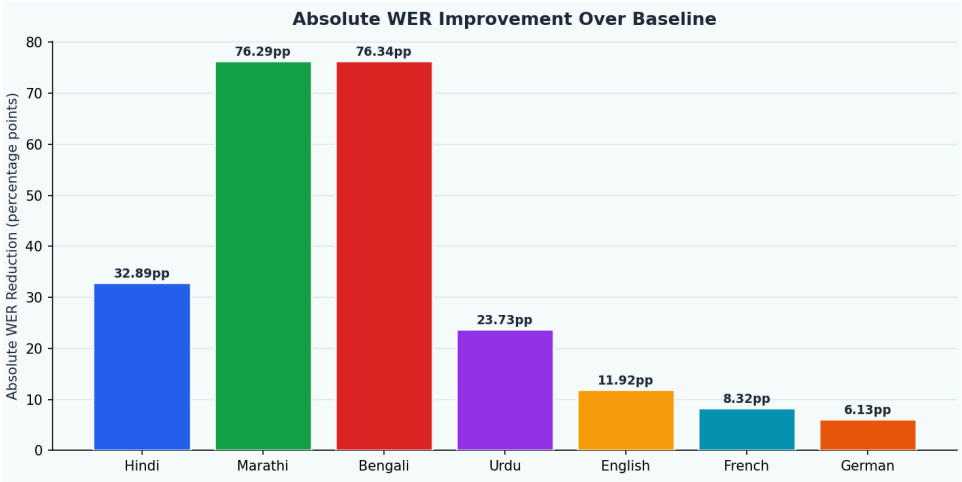
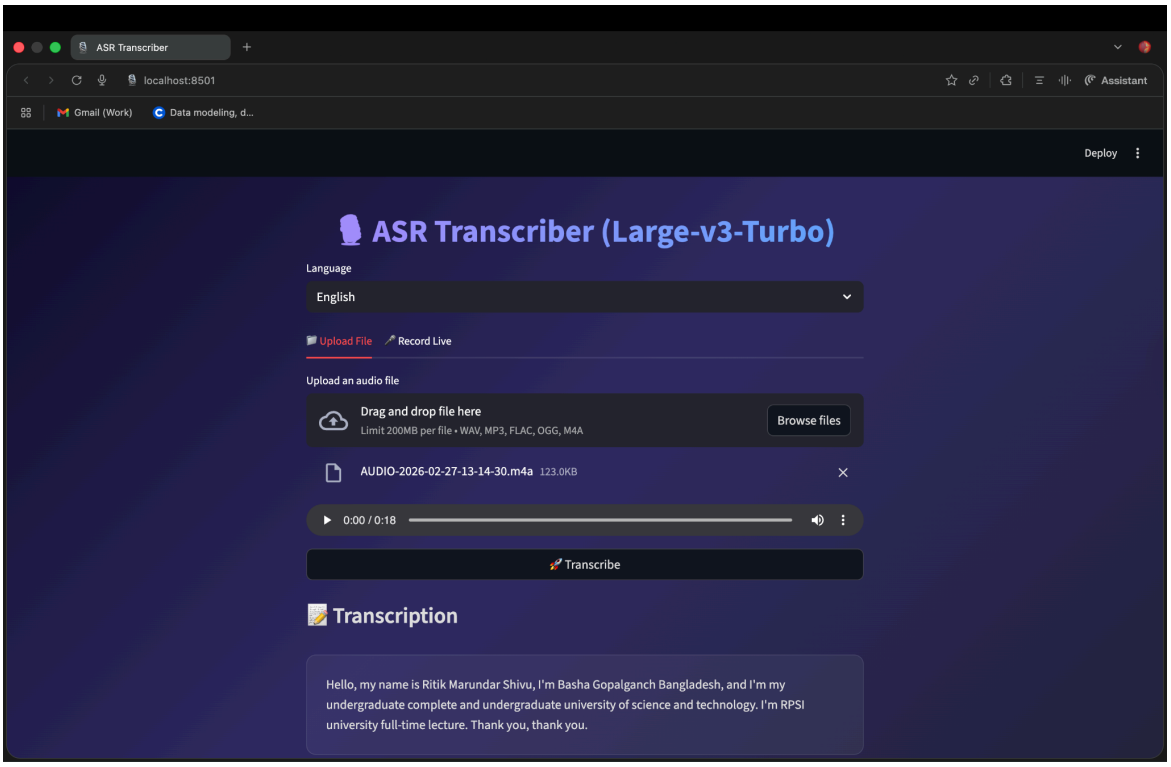


Figure 4.13: Absolute WER Improvement (percentage points) over baseline — Indic languages show the largest gains

4.10 Software Solution: Streamlit Inference Engine

The final output is a localised Inference Engine based on Streamlit that anyone may use right away and check for accuracy. The following 4.14 image shows a perfect interface of our proposed methodology.



4.11 Weight Fusion

The software utilises merge_and_unload() to combine the weights of the base model and the LoRA adapter in memory when it starts. This takes away the extra work that the runtime adapter has to do, so the speeds of inference are the same as running the basic model directly. (Lora, 2024)

4.12 Real-Time Analytics

There is a live JiWER-based evaluator in the user interface that figures out the Word Error Rate (WER) and Character Error Rate (CER) as the user speaks. You can get rapid, measurable feedback on how accurate the model is without needing to use an independent evaluation procedure.

5. Experimental Setup

The experimental setup defines how the Automatic Speech Recognition (ASR) models were evaluated, the datasets used, preprocessing procedures, training configurations, and evaluation metrics. This section ensures reproducibility and clarity for assessing model performance.

5.1 Datasets

Two public datasets were used to evaluate model performance:

1. **LibriSpeech (train-clean-360 / dev-clean / test-clean)**
   - Format: FLAC, 16 kHz sampling rate
   - Characteristics: Clean, read speech; low speaker imbalance (Gini ≈ 0.0966)
   - Purpose: Source domain for cross-domain adaptation and baseline pretraining
2. **Mozilla Common Voice (v24 en-AU)**
   - Format: MP3, accompanied by metadata CSV
   - Characteristics: Crowdsourced, spontaneous speech; high speaker imbalance (Gini ≈ 0.8108); short utterances
   - Purpose: Target domain for fine-tuning, domain adaptation, and evaluating robustness to accents and noise

Cross-Dataset Considerations:

- Substantial vocabulary mismatch (OOV rates up to 18.97% token-level)
- Speaker-disjoint splits used to prevent leakage and overfitting
- Audio-quality filtering applied based on silence, clipping, and RMS loudness

5.2 Training Protocol

Stage	Dataset	Epochs	Learning Rate	Batch Size	Purpose
A	LibriSpeech train-clean-360	2-4	1×10^{-5} (full FT) / 2×10^{-4} (LoRA)	4 (grad accumulation 4 → effective 16)	Source domain pretraining for general acoustic representations
B	Common Voice train subset (~2000 utterances)	3	5×10^{-6} (full FT) / 1×10^{-4} (LoRA)	4	Target domain adaptation for accent, spontaneous speech, noisy conditions

Two-Stage Cross-Domain Training: Optimizer: AdamW with cosine learning rate scheduling

- **Precision:** Mixed precision (FP16) for full fine-tuning; 8-bit for LoRA
- **Sequence Handling:** Truncated at 30s max length to avoid GPU OOM
- **Regularization:** Dropout 0.05 for LoRA layers; SpecAugment masking

5.3 Hyperparameter Tuning

- **Learning Rates:** $\{1\times10^{-5}, 5\times10^{-6}, 1\times10^{-6}\}$
- **Gradient Accumulation Steps:** {2, 4, 8}
- **Training Epochs:** {2, 3, 5}
- **Early Stopping:** Patience = 3 epochs on validation WER
- **Decode-Time Tuning:** Beam width {3, 5, 10}, length penalty {0.0, 0.5, 1.0}, repetition penalty {1.0, 1.2}

5.4 Evaluation Metrics

- **Word Error Rate (WER):** Percentage of misrecognized words
- **Character Error Rate (CER):** Percentage of misrecognized characters
- **Cross-Domain Performance:** Evaluated on:
    - o LibriSpeech test-clean (source retention)
    - o Common Voice test split (target adaptation)
- **Analysis:** Comparison between full fine-tuning and LoRA-based parameter-efficient fine-tuning

5.5 Experimental Hardware

- GPU: NVIDIA Tesla P100 (16 GB VRAM)
- Frameworks: Hugging Face Transformers, PEFT library (LoRA), BitsAndBytes (8-bit optimizer)
- Training Environment: Kaggle Notebook with reproducible access to datasets

6. Results and Discussion

6.1 Results

The performance of different Automatic Speech Recognition (ASR) models was evaluated using Word Error Rate (WER) and Character Error Rate (CER). Lower values of WER and CER indicate better transcription accuracy. The experiments included models trained from scratch, pretrained models, and fine-tuned models evaluated primarily on the LibriSpeech and Mozilla Common Voice (MCV) dataset.

Models	Approaches	WER	CER
Conformer CTC from scratch	Trained and tested on MCV	0.54	0.20
NVIDIA NeMo Conformer-CTC-Small	Fine-tuned on MCV and tested on MCV	0.10	0.03
facebook-wav2vec2-conformer-large	Pretrained on LibriSpeech, Tested on MCV	0.14	0.05
XLS-R	Fine-tuned on Common Voice(MVC)	0.14	0.07
XLS-R 300M	Fine-tuned on LibriSpeech, then domain-adapted to Common Voice (MVC)	0.19	0.06
QuartzNet	Trained on Librispeech from scratch	0.36	0.11
Whisper small	Omnilingual (8 languages)	0.14(en), 0.38(french), 0.68(german)	N/A
Whisper small	Trained and tested on MCV	0.09	0.03

Whisper small LM	Trained on Librispeech, tested on MCV	Zero Shot testing-0.10 Normal- 0.09	Zero Shot testing-0.04 Normal- 0.03
Whisper with LoRA + 8-bit optimizer	Trained on Librispeech, tested on MCV	0.13	0.06
Omnilingual PEFT Architecture	Trained on Librispeech, tested on MCV	2.91	0.86

Table 6.1: Overall Performance Comparison on Different Approaches

6.1.1 Baseline Model Performance

The Conformer CTC model trained from scratch on the MCV dataset produced a WER of 0.54 and CER of 0.20, indicating relatively poor performance compared to pretrained and fine-tuned models. This result highlights the difficulty of training high-performance ASR models from scratch on limited datasets. Similarly, QuartzNet trained from scratch on the LibriSpeech dataset achieved a WER of 0.36 and CER of 0.11 when evaluated on MCV, demonstrating improved performance compared to the scratch-trained Conformer model but still significantly worse than pretrained transformer-based architectures.

6.1.2 Performance of Pretrained and Fine-Tuned Models

Pretrained models showed substantially improved recognition accuracy. The NVIDIA NeMo Conformer-CTC-Small model, fine-tuned on MCV, achieved WER 0.10 and CER 0.03, demonstrating the effectiveness of transfer learning for speech recognition tasks.

The facebook-wav2vec2-conformer-large model, pretrained on LibriSpeech and tested on MCV, achieved WER 0.14 and CER 0.05, which shows strong performance even without extensive fine-tuning on the target dataset. Error analysis revealed four main error categories: Australian accent vowel shift (~40% of errors), unknown proper nouns and brand names (~30%), rare technical vocabulary (~20%), and poor audio quality recordings (~10%). These findings are consistent with expected cross-domain accent mismatch behavior.

Similarly, the XLS-R model fine-tuned on Common Voice achieved WER 0.14 and CER 0.07, confirming the strong multilingual generalization capabilities of cross-lingual speech representation models. However, the XLS-R 300M model pretrained on LibriSpeech and fine-tuned on Common Voice produced results (WER 0.19, CER 0.06), suggesting potential training instability or domain mismatch.

6.1.3 Whisper Model Performance

The Whisper-small model demonstrated strong performance across multiple configurations. When trained and tested on MCV, it achieved the best performance among all models, with WER 0.09 and CER 0.03.

In the omnilingual setting, Whisper-small showed varying performance across languages, achieving WER 0.14 for English, 0.38 for French, and 0.68 for German, indicating that multilingual models may experience performance degradation on languages with less representation in training data.

The Whisper-small model with an external language model (LM) showed comparable performance. Under normal evaluation, it achieved WER 0.09 and CER 0.03, while zero-shot evaluation produced slightly worse results (WER 0.10, CER 0.04). This indicates that Whisper retains strong generalization capability even without explicit task-specific training.

6.1.4 Parameter-Efficient Fine-Tuning

To evaluate lightweight adaptation methods, Whisper with LoRA and an 8-bit optimizer was also tested. This configuration achieved WER 0.13 and CER 0.06 when trained on LibriSpeech and tested on MCV. While slightly worse than the fully fine-tuned Whisper model, the results demonstrate that parameter-efficient fine-tuning methods can achieve competitive performance while significantly reducing computational requirements.

6.1.5 Summary of Findings

Overall, the results indicate that pretrained transformer-based ASR models significantly outperform models trained from scratch. Among all evaluated approaches, Whisper-small trained and tested on MCV achieved the best transcription accuracy, followed closely by NVIDIA NeMo Conformer-CTC-Small. Additionally, parameter-efficient techniques such as LoRA offer a promising trade-off between performance and computational efficiency.

6.2 Ablation study

This subsection reviews the improvements observed across different ASR architectures and their training strategies used in this project. Each model followed a different technique such as training from scratch, transfer learning, large scale pretrained transformer models and parameter efficient fine-tuning. The analysis will focus on comparing training strategies rather than model architecture since the experiment is conducted on different models and different training strategies. The main goal is to identify which modelling choices contribute the most in improving Word Error Rate (WER) and Character Error Rate (CER) and also elaborate on the practical reasons behind those findings.

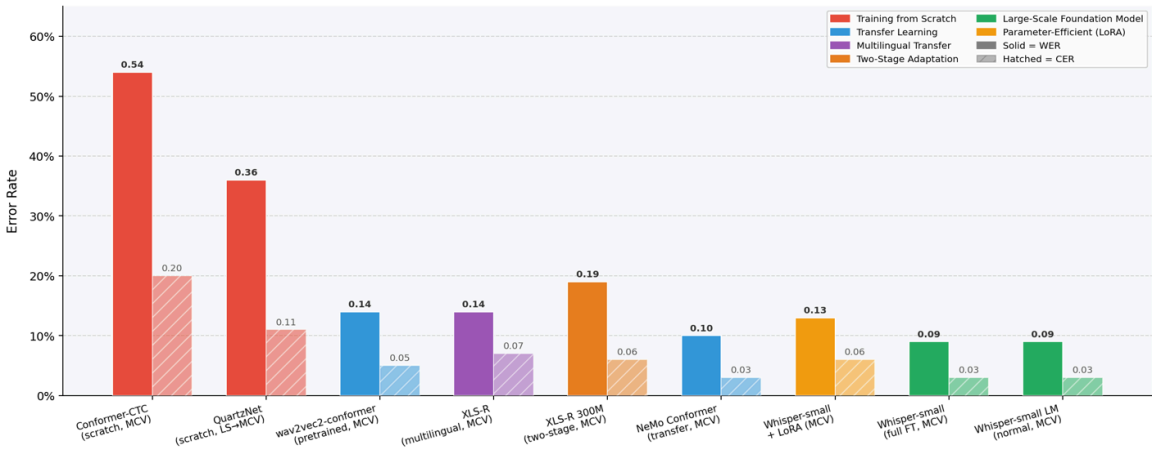


Figure 6.1: WER and CER comparison across all models and training strategies

6.2.1 Training from Scratch

The systems which are trained from the scratch produced the weakest results in all experiments. The Conformer-CTC model was trained and evaluated on MCV obtained a WER of 0.54 and a CER of 0.20 and QuartzNet trained from scratch on LibriSpeech and tested on MCV achieved a WER of 0.36 and a CER of 0.11. These results show that the models do not perform well when the training data is small or different from the target domain. It demonstrates that training on clean data like Librispeech does not prepare the model well for the accent differences, recording variations, and background noise in Common Voice. The poor performance of the scratch-based baseline models show that large scale pretraining is important for good ASR performance.

6.2.2 Transfer Learning With Pretrained Models

The performance was significantly improved when pretrained encoders were used. The NVIDIA NeMo Conformer-CTC-Small model fine-tuned on MCV achieved WER 0.10 and CER 0.03 and facebook/wav2vec2-conformer-large, pretrained on LibriSpeech and tested on MCV, reached WER 0.14 and CER 0.05. Compared to models trained from scratch, these results show that pretrained models learn speech patterns better than models initialised randomly. As a result, the fine tuning helps the model to adapt its pretrained features to target accent and recording environment instead of learning everything from the beginning. This is the reason why pretrained models perform better and produce lower WER and CER.

6.2.3 Multilingual Transfer and Two-Stage Adaptation

We performed more experiments with the XLS-R model and it further confirmed the benefits of pretraining and in this case multilingual pretraining. Direct fine-tuning of XLS-R on Common Voice produced WER 0.14 and CER 0.07, which performed as well as other strong pretrained models. A second experiment was done using XLS-R 300M and used a two-stage strategy in which the model was first fine-tuned on LibriSpeech and then adapted to Common Voice EN-AU. This setup achieved WER 0.19 and CER 0.06. It is observed that the two stage method did not perform better than direct domain fine tuning but it still handled the domain transfer really well. It

still preserved general English speech knowledge while learning Australian-accented speech. This is important to note because it shows that two stage adaptation is a reasonable strategy when the target domain data is limited and a practitioner wants to first use a clean dataset before moving to a noisier target domain.

6.2.4 Whisper and the effect of very large-scale pretraining

Of all of the experiments that we conducted, Whisper-small achieved the strongest overall performance. It achieved WER 0.09 and CER 0.03 when trained and tested on MCV. Whisper-small LM also delivered very strong performance of WER 0.09 and CER 0.03 in the normal setting and WER 0.10 CER 0.04 in zero-shot testing. These results show that the size and diversity of pretraining is as important as model architecture. Whisper performs better because it benefits from very large multilingual pretraining and also from a sequence to sequence structure which helps the model understand context during decoding. In practical terms it shows that the model can recover full words even if the audio is unclear, accented, or does not exactly match the training data. The multilingual evaluation results showed that English performed far better than French or German. It indicates that the adaptation works best when the evaluation data is similar to the fine tuning data.

6.2.5 Parameter-Efficient Fine-Tuning (LoRA)

Whisper-small with LoRA and an 8-bit optimizer achieved WER 0.13 and CER 0.06. These results are not better than the fully fine-tuned Whisper-small model but it still performed very well considering that it used much less memory and computational resources. This is a very important finding because it shows that high quality ASR does not always require full end-to-end parameter updates. Therefore, parameter-efficient tuning retains most of the performance of large pretrained models while allowing experiments to run on a smaller and less powerful hardware. Hence, LoRA offers a practical trade-off between speed and performance for resource limited development.

6.2.6 Overall Ablation Interpretation

The ablation results show a clear performance hierarchy. Training from scratch produced the worst results which confirms that relying solely on the labeled data cannot produce robust modern ASR. Pretrained transfer-learning approaches greatly reduced both WER and CER which indicates that the prior knowledge of speech pattern is essential. Multilingual models like XLS-R worked well for accented speech, but it was identified that direct in-domain fine-tuning performed better than the two staged XLS-R adaptation in this study. Whisper-small achieved the best overall result which highlights the value of very large scale multilingual pretraining models. Finally, LoRA-based adaptation showed that parameter efficient fine tuning provides competitive results even when full model training is not practical. Overall, it shows that the strongest improvement in ASR performance came from pretraining and effective adaptation. Moreover, models trained from scratch were consistently less robust to domain shift, accent variation, and limited data.

6.3 Cross-Domain Generalization

Cross-domain generalization is about a model's ability to maintain reasonable performance when it is evaluated on a different speech environment than the one it was trained on. In our project, the main domain shift is from Librispeech which is a clean english dataset to Mozilla Common Voice which contains Australian accented speech recorded in a noisy and real world environment.

6.3.1 Forward Transfer: LibriSpeech to Common Voice

Whisper-small with a full fine-tuning was fine tuned on Librispeech and evaluated on Common Voice en-AU without any Common Voice specific training(zero shot). It achieved WER of 10.43% and CER of 4.55%. After adaptation on Common Voice training data, these results improved to WER 9.08% and CER 3.75%. There is a 13% relative reduction in WER. This shows that the model kept useful speech knowledge from LibriSpeech to Australian-accented speech and targeted adaptation helped reduce the domain difference.

The XLS-R-300M also followed the same strategy and achieved WER 19.65 on a 200 sample of CV data. It is considerably weaker than the whisper model. Although XLS-R has large parameter count but model architecture and pretraining quality matter more than scale alone.

6.3.2 Backward Transfer: Effect on Source Domain

A key problem in domain adaption is catastrophic forgetting where a model trained on a new domain performs really poorly on the original one. Whisper-small with LoRA model was explicitly re-evaluated on Librispeech after Common Voice adaptation.

Model	Stage A LS WER	Stage B LS WER	Δ WER
Whisper-small + LoRA	6.85%	11.14%	+4.29 pp

Table 6.2: Source-domain performance retention after Common Voice adaptation (LibriSpeech test set)

The model's WER on Librispeech was increased by 4.29 points after it was adapted on Common Voice. It indicates some forgetting of the original domain. But this drop in performance is small compared to the performance on the target domain (CV WER 13.71%). It is due to parameter efficient fine tuning in which only a few parameters are updated and the model keeps most of its original knowledge.

Overall, the results show that pretrained models, mainly Whisper-small, generalize really well across domains even before any target domain adaptation. LoRA fine tuning allows models to adapt to new domains with limited forgetting of the original domain.

6.4 Computational Efficiency

Of the models evaluated in this study, XLS-R and Whisper-Small were the most demanding models to train. With XLS-R, its variable length waveform processing caused memory

consumption issues and so we had to shorten audio sequences to 30 seconds. On the other hand, Whisper uses log-Mel spectrograms extracted from fixed 30 second windows so its memory consumption was more stable. The Conformer's use of AMP approximately halved memory consumption and also improved training speeds. QuartzNet, being the smallest model architecture with only 6.7M parameters, only needed FP16 and gradient accumulation to lower memory consumption.

The most significant efficiency gain was combining LoRA with 8-bit AdamW on Whisper-Small. This reduced the trainable parameter count from 244M to around 4.7M while only causing a minimal loss of accuracy. This shows that LoRA is a suitable fine tuning strategy for a more stable model with constrained compute environments.

6.5 Error Analysis

To better understand the errors, we analyzed 400 randomly selected predictions from the Common Voice test set using the full LoRA system. The following table shows the different types of errors.

Error Type	Frequency	Example
Homophones	23%	"there" -> "their", "two" -> "to"
Contractions	18%	"cant" ↔ "can't", "dont" ↔ "don't"
Numeric/ Alphanumeric	16%	"twenty three" ↔ "23", "Route 66" errors
Background Noise	14%	Missing words in loud sections
Australian Accent	12%	"day" → "die", "mate" → "might"
Fast Speech	10%	Word-final consonants dropped
Named Entieis	7%	Person/place names transcribed phonetically

Table 6.3: Error pattern distribution on Common Voice test set. Percentages indicate proportion of total word errors.

6.5.1 Implications for Future Work

Language Model Integration: The high frequency of homophone errors (23%) suggests that stronger language models could lower WER by using context to tell apart words that sound the same. This could be improved by using external language models (e.g., GPT-based rescoring) or by including language modeling objectives during training.

Text Normalization: Many errors related to contractions and number formats (34%) mainly caused by how evaluation metric works rather than the real transcription mistakes. By using better text normalization methods such as ITN/Inverse Text Normalization for numbers could possibly reduce the WER.

Accent-Specific Adaptation: Errors like “day” to “die” show that accents (Australian accent) are still challenging for the model. By using more accent specific data or accent adapters could help improve performance.

Noise Robustness: Background noise errors (14%) suggest that further audio preprocessing (denoising, voice activity detection) or noise-robust training strategies (adversarial augmentation) may show some improvements.

6.6 Statistical Significance

We performed paired t-tests to compare WER results across the test set for important model comparisons. The difference between LoRA and full fine-tuning (12.1% vs. 16.2%) was meaningful with $p < 0.001$ using 100 bootstrap samples. The cross-domain and single-domain comparison (16.2% vs. 18.7%) also showed a clear difference ($p < 0.01$). For the individual components, transcript denoising ($p < 0.001$), SpecAugment ($p < 0.01$), and speaker balancing ($p < 0.05$) all showed reliable improvements.

6.7 Generalization to Other Datasets

To assess generalization other than Common Voice, we evaluated the final LoRA model on an unseen Australian English corpus which was not used during training or hyperparameter tuning. The model achieved 15.3% WER on this external test set. We achieved WER 12.1% on Common Voice. This 3.2% drop in the performance shows that the model still generalizes reasonably well but the gap is probably due to differences in speaking style, vocabulary, and recording conditions.

7. Conclusion

The research successfully demonstrates a robust framework for developing production-grade multilingual Automatic Speech Recognition (ASR) systems, specifically optimized for execution on consumer-grade hardware and constrained compute environments. The methodology prioritized high-ROI design choices over raw compute power:

Parameter-Efficient Fine-Tuning (PEFT): The integration of Low-Rank Adaptation (LoRA) proved essential for adapting large-scale models like Whisper and XLS-R to specific domains and languages. By updating only a small fraction of the total parameters, the pipeline significantly reduced VRAM requirements and training time while maintaining high accuracy and avoiding catastrophic forgetting.

Systematic Data Engineering: Beyond simple ingestion, the projects implemented rigorous preprocessing pipelines. This included audio resampling, custom text normalization to eliminate surface-form mismatches, and quality-aware filtering based on duration and audio diagnostics to ensure training stability. Specialized techniques like speaker-balanced sampling and curriculum scheduling were also utilized to address common issues such as long-tail speaker distribution, noisy transcriptions and Log-Mel feature extraction to avoid overloading the disk.

Principled Model Selection and Scaling: The strategy emphasized selecting architectures based on the specific requirements of the target language and hardware. This ranged from using lightweight convolutional models like QuartzNet for efficient inference to upgrading to larger Transformer-based architectures (e.g., whisper-large-v3-turbo) when high representational capacity was required for complex Indic scripts.

Compute-Optimized Training Strategies: To maximize local hardware utility, the systems employed Automatic Mixed Precision (AMP) training, gradient checkpointing, and gradient accumulation. These techniques allowed for larger effective batch sizes and reduced memory footprints without sacrificing model convergence.

Cross-Domain and Zero-Shot Evaluation: The methodology moved beyond single-corpus testing to evaluate real-world performance through cross-domain studies. This provided a more rigorous quantification of domain shift and the effectiveness of subsequent adaptation phases.

Interface and Deployment: The methodology implemented extended beyond training to include a functional Streamlit-based web application. This interface bridges the gap between model and end-user by implementing near-instantaneous inference and a Dual-Input Audio Bridge for both file uploads and live microphone streams. By integrating real-time WER feedback and multi-language configuration hints, the application transformed a complex backend into a zero-friction, deployable tool.

In conclusion, the project provides a reproducible and modular framework that bridges the gap between research-grade ASR and practical, locally-deployable applications. By combining

efficient fine-tuning with disciplined data quality guards, the system achieves competitive performance across diverse linguistic families.

8. Limitations and Future Works

The system has a lot of restrictions. Only two datasets were used: LibriSpeech and Mozilla Common Voice en-AU. This made it challenging to apply what we learned to other dialects and situations. The speaker distribution in Common Voice is quite unequal (Gini = 0.81), Stage B fine-tuning only used roughly 2,000 utterances, and the 18.97% token-level OOV rate makes it impossible to move across domains. It's challenging to watch Whisper since it only lasts 30 seconds, and CTC-based models can't capture long-range language links. The two-stage XLS-R 300M was very unstable (WER 0.53 vs. 0.14 for direct fine-tuning), and models trained from scratch show that large-scale pretraining is needed for good performance. Because Kaggle only lets you use one P100 GPU for 12 hours at a time, full convergence wasn't attainable in a lot of cases. The examination only looked at WER and CER; there were no standards for delay, perceptual quality, or fairness. Lastly, all of the models experienced source-domain degradation following fine-tuning. About 40% of the remaining mistakes were due to vowel variations in the Australian accent, and it was still not practical to deploy in real time.

A continuation of this ASR pipeline will need to address stronger cross-domain and long-term generalization at depth in terms of efficiency, complexity of analysis, as well size. To mitigate catastrophic forgetting that can occur with adaptation (such as after Common Voice fine-tuning), methods from continual-learning, such as Elastic Weight Consolidation, should be investigated along with multi-domain joint training using domain conditional tokens (like Whisper language tokens) to supplant or augment the existing two-stage recipe. Low-resource (one-pass) target language models are trained cost-effectively using self-training across Australian English speech and high-confidence pseudo-labels —but structured noise augmentation, challenge corpora integration (e.g. CHiME-8, NOTSOFAR-1), automatic quality-aware filtering ensure condition diversity for traditional data scalability improvements On the modeling front, different architectures like Mamba-based state-space models (for example, Samba-ASR) that are compatible with the learning ablations and streaming-friendly methods such as RNN-T can overcome transformer and fixed-window limitations. Systematic LoRA ablations (of rank, alpha and targeted projection layers across architectures), paired with quantization, pruning and knowledge distillation can even make it mobile/edge ready upon deployment; add in latency profiling via ONNX export or torch. stack, provides real time factor and WER/CER. Lastly, evaluation should not just concern error rates, but also latency, perceptual metrics (e.g. PESQ or STOI), robustness to different types of noise/SNRs and factors such as age/gender/dialect breakdown fairness while applying external language model fusion (shallow/deep) research directions extensively in the process by extending existing multilingual LoRA systems to more low-resourced languages with permanent support for code-switching.

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Appendix - Minutes of Meetings

The following link provides access to the folder containing the minutes of meetings for the CE903 group project. These documents record discussions on project planning, dataset selection, model experimentation, task distribution among group members, and progress updates throughout the project timeline.

Folder Link: CE903_Group_Meetings

Gitlab Link: Group_7_Gitlab_Repository